51st AIAA Aerospace Sciences Meeting
7-10 January 2013, Grapevine, TX

APA-24 Low Boom CFD and Experimental Results

Near-Field Pressure Measurement of Several Models in JAXA's 1m x 1m Supersonic Wind Tunnel

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Outline

- Supersonic Research Program at JAXA
- Objectives
- Supersonic Wind Tunnel Test in 2011
- Supersonic Wind Tunnel Test in 2012
- Summary

Supersonic Research Program at JAXA

National EXp. Supersonic Transport Program

NEXST program

Silent Supersonic Tech. Research Program

S-cube Program

Year

1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014~

Design concept

- **Cranked arrow wing**
- **Warp airfoil**
- **Area-ruled fuselage**
- **Natural Laminar flow**

Design tools

- **Carlson warp**
- **CFD-based inverse design**

Flight demo.

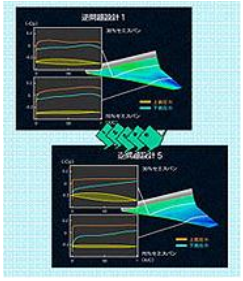
Design concept

- **Non-axisymmetric nose**
- **Lifting aft-fuselage**
- **Inversely cambered stabilizer**

•Design tools

- **CFD-based optimization**
- **MDO**

Flight demo.



NEXST-1



Non-powered

Rocket-boosted

Phase 1 (D-SEND) | Phase 2 (S3TD)

D-SEND#1 **D-SEND#2** **S3TD**







Non-powered

Balloon drop

Jet-powered

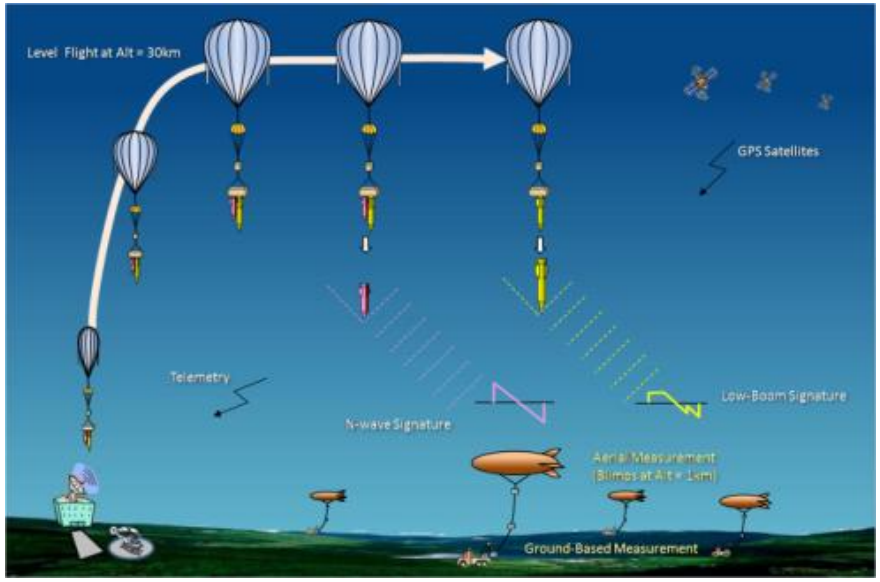
Takeoff/Land

Low-drag tech. R&D → **Low-sonic-boom tech. R&D**

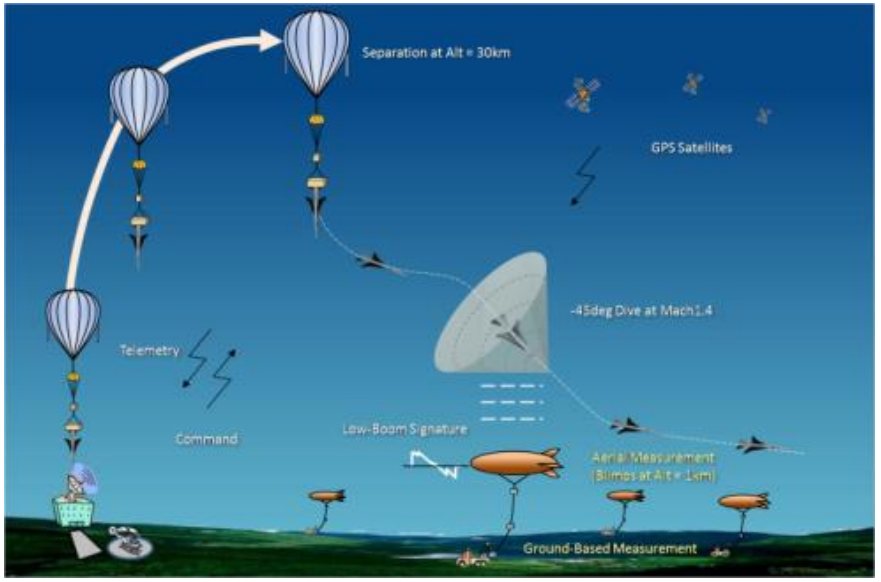
"D-SEND" Project

Drop test for **S**implified **E**valuation of **N**on-symmetrically **D**istributed sonic boom

- Objective
  - To validate JAXA's low-boom design concepts.
  - To establish airborne sonic boom measurement system.

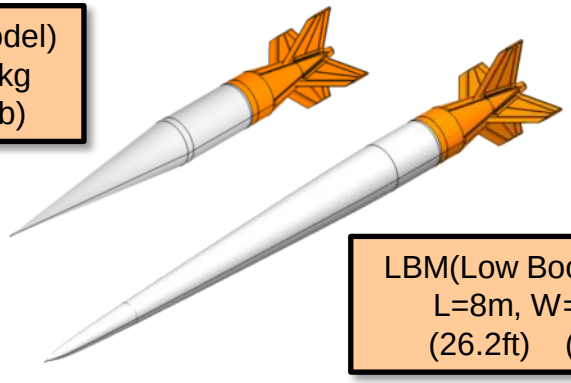


D-SEND#1 (2011.05)



D-SEND#2 (2013)

NWM（N Wave Model)
L=5.6m, W=700kg
(18.4ft)　(1.5klb)



LBM(Low Boom Model)
L=8m, W=630kg
(26.2ft)　(1.4klb)



S3CM(S-cube Concept Model)
L=7.7m, W=1000kg
(25.3ft)　(2,2klb)

D-SEND#1

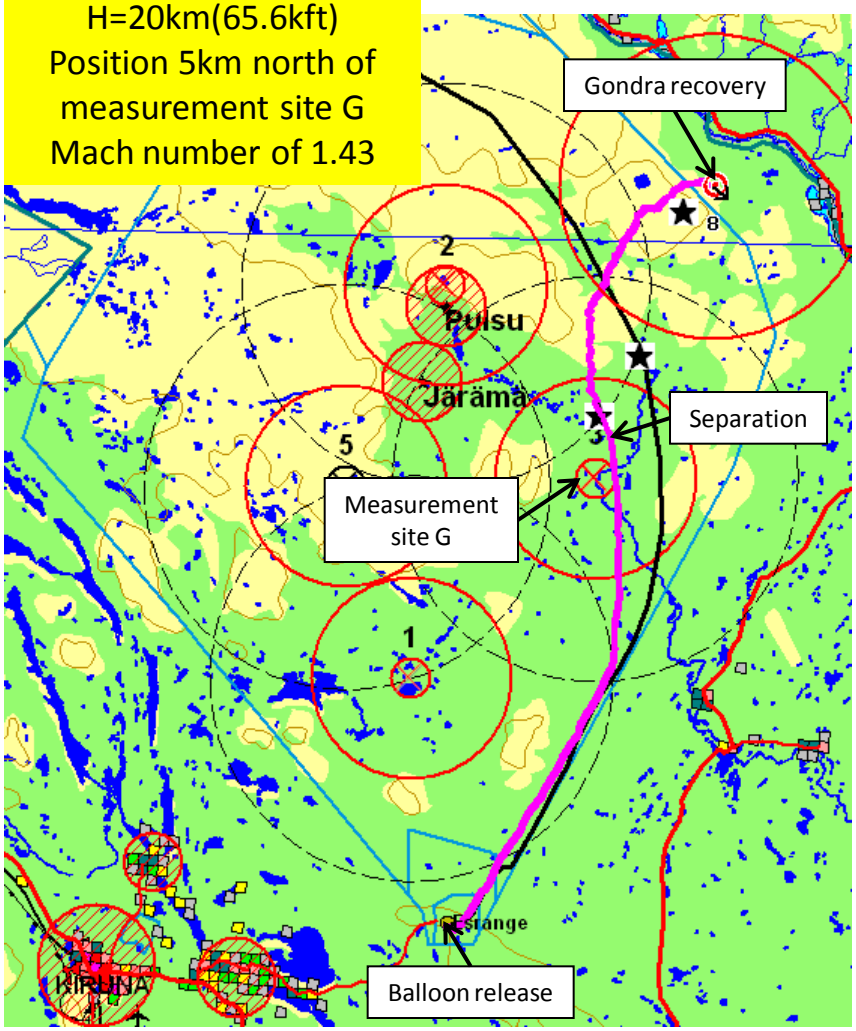
- Objectives
  - To validate the low-boom demonstration technique.
  - To establish airborne boom measurement system.
- Overview
  - Date: May 7th and 16th, 2011
  - Place: Sweden (NEAT)
  - # of Drop Tests: 2
  - # of Drop Bodies: 2 in each test
    - N-Wave Model (NWM)
    - Low-Boom Model (LBM)



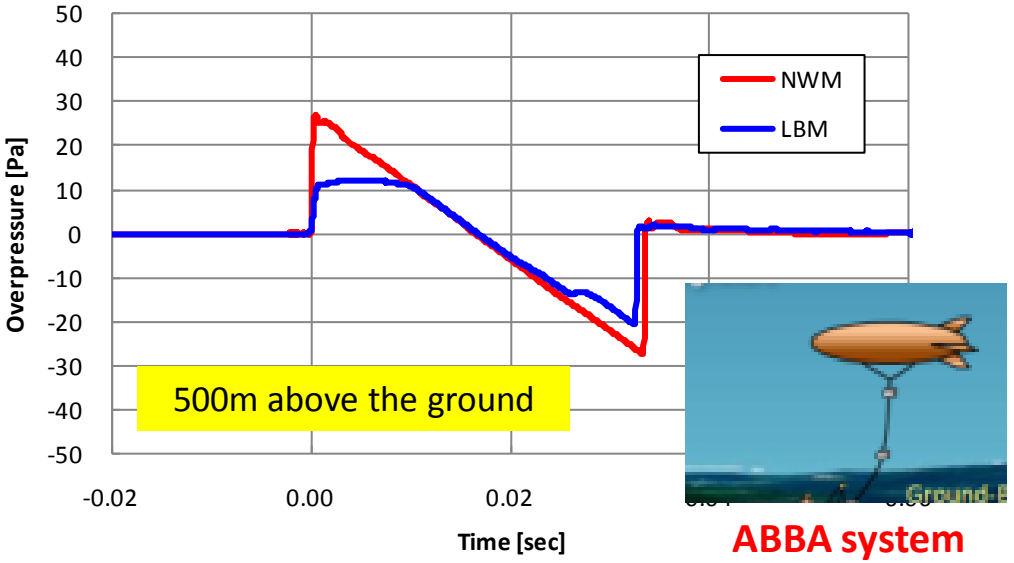


D-SEND#1 1st Drop Test

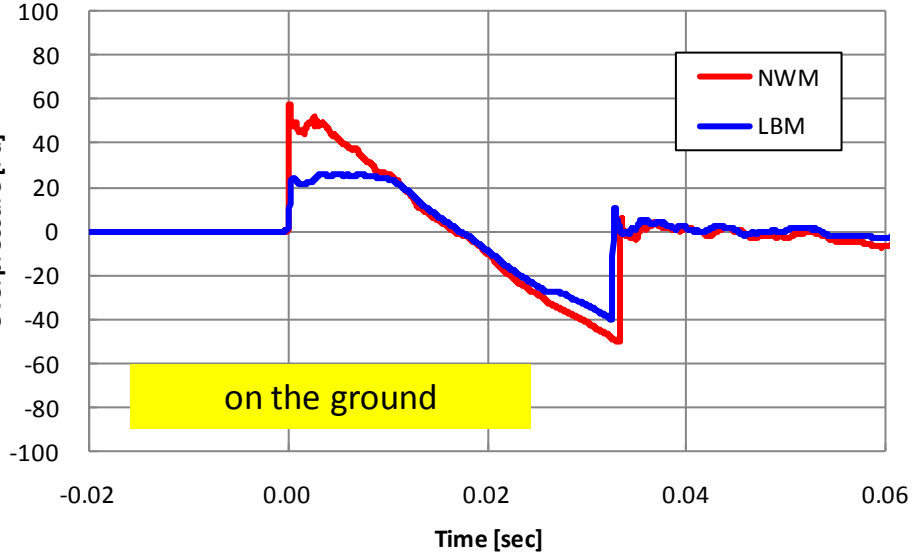
Separated from
H=20km(65.6kft)
Position 5km north of
measurement site G
Mach number of 1.43



Balloon Trajectory



ABBA system



D-SEND Database

Drop test for Simplified Evaluation of Non-symmetrically Distributed sonic boom

D-SEND Database

Japanese / English



HOME | About D-SEND | D-SEND#1 | D-SEND#2 | ABBA Test | Contact

D-SEND Experimental Data



- To acquire the airborne boom measurement technique.
- To make sure the possibility of the low-sonic-boom design concept validation with a scaled model.
- To prepare the D-SEND#2 flight test (Test protocol and measurement system check)



- To demonstrate both front and rear shock shaping design.
- To acquire the low-boom sonic-boom signature measurement technique.
- To obtain the validation data for low-boom sonic-boom signature propagation tools.



- To establish sonic-boom measurement technique (airborne, on the ground).
- To gather the data for sonic-boom evaluation.
- To make sure the validity of the flight test in the D-SEND project.
- To obtain the validation data for sonic-boom prediction method.

http://d-send.jaxa.jp/d_send_e/index.html

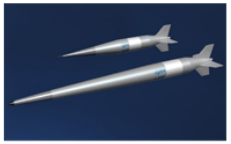
HOME | About D-SEND | D-SEND#1 | D-SEND#2 | ABBA Test | Contact

HOME > D-SEND#1

D-SEND#1

D-SEND#1 Overview

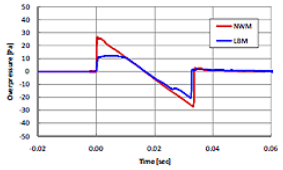




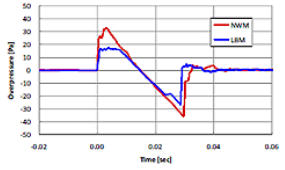
JAXA conducted the D-SEND#1 drop tests on May 7th and 16th in 2011 at Esrange test range in Sweden supported by the Swedish Space Corporation(SSC). The low sonic-boom design results have been validated with an axisymmetric body for the first time.

The D-SEND Database provides the geometries of two D-SEND#1 models named NWM(N-wave Model) and LBM(Low-Boom Model) as well as these drop profile data and sonic boom data obtained in both drop tests.

	1st Drop Test	2nd Drop Test
Balloon Release Time	AM5:44 7th May,2011 LT	AM5:30 16th May,2011 LT
Separation Time	AM7:02 7th May,2011 LT	AM7:36 16th May,2011 LT
Separation Altitude	About 21km	About 27km
Maximum Mach Number	About M=1.4	About M=1.7



1st Drop Test



2nd Drop Test

Data Download

D-SEND#1

Model Geometries

- NWM/LBM configuration data (D-SEND#1_Model.lzh)

1st Drop Test

- Drop profile/Weather data (D-SEND#1-1_PathWeather.lzh)
- Sonic-boom data (D-SEND#1-1_Boom.lzh)

2nd Drop Test

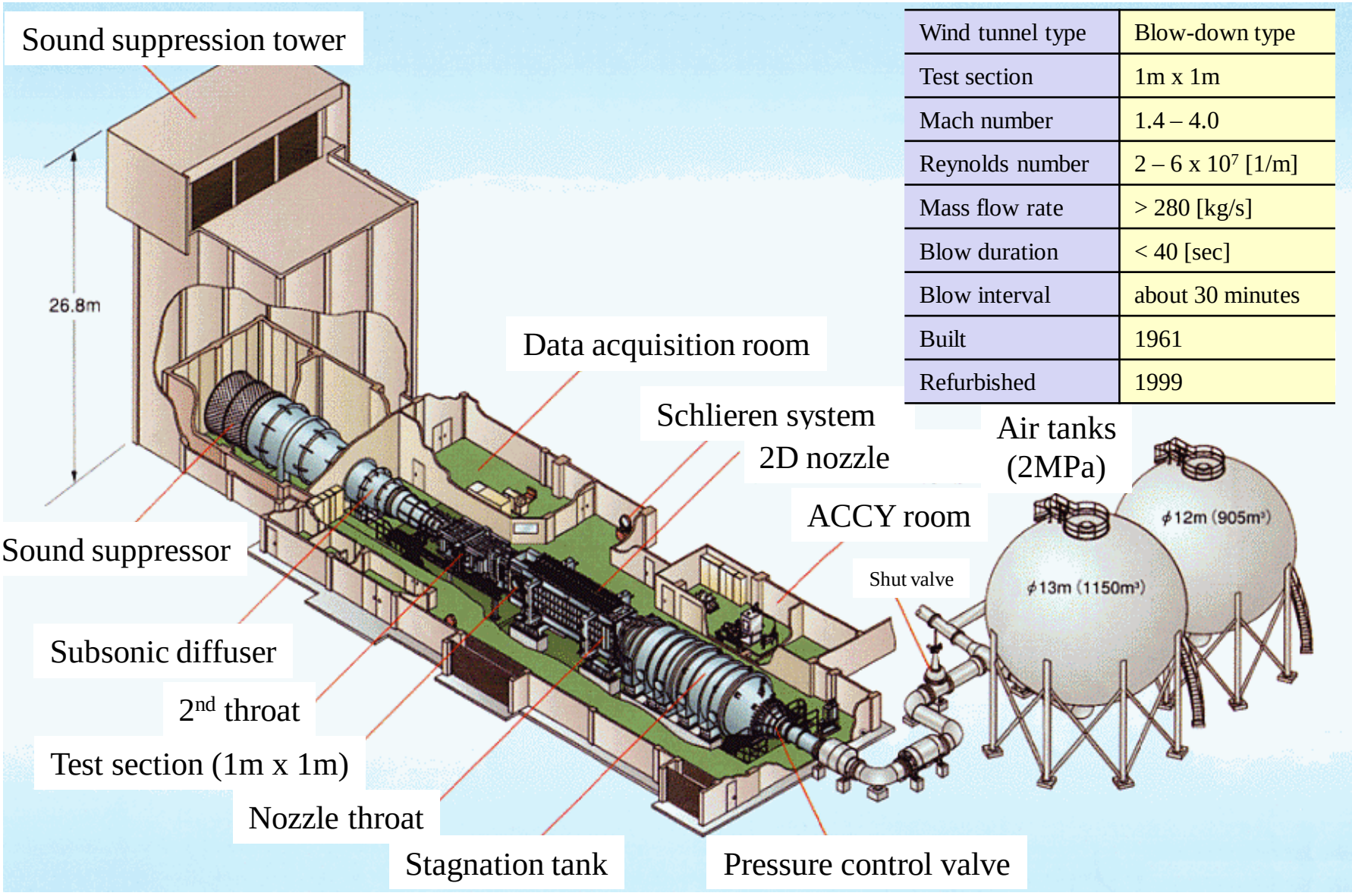
- Drop profile/Weather data (D-SEND#1-2_PathWeather.lzh)
- Sonic-boom data (D-SEND#1-2_Boom.lzh)

Supersonic Wind Tunnel Test

Objectives

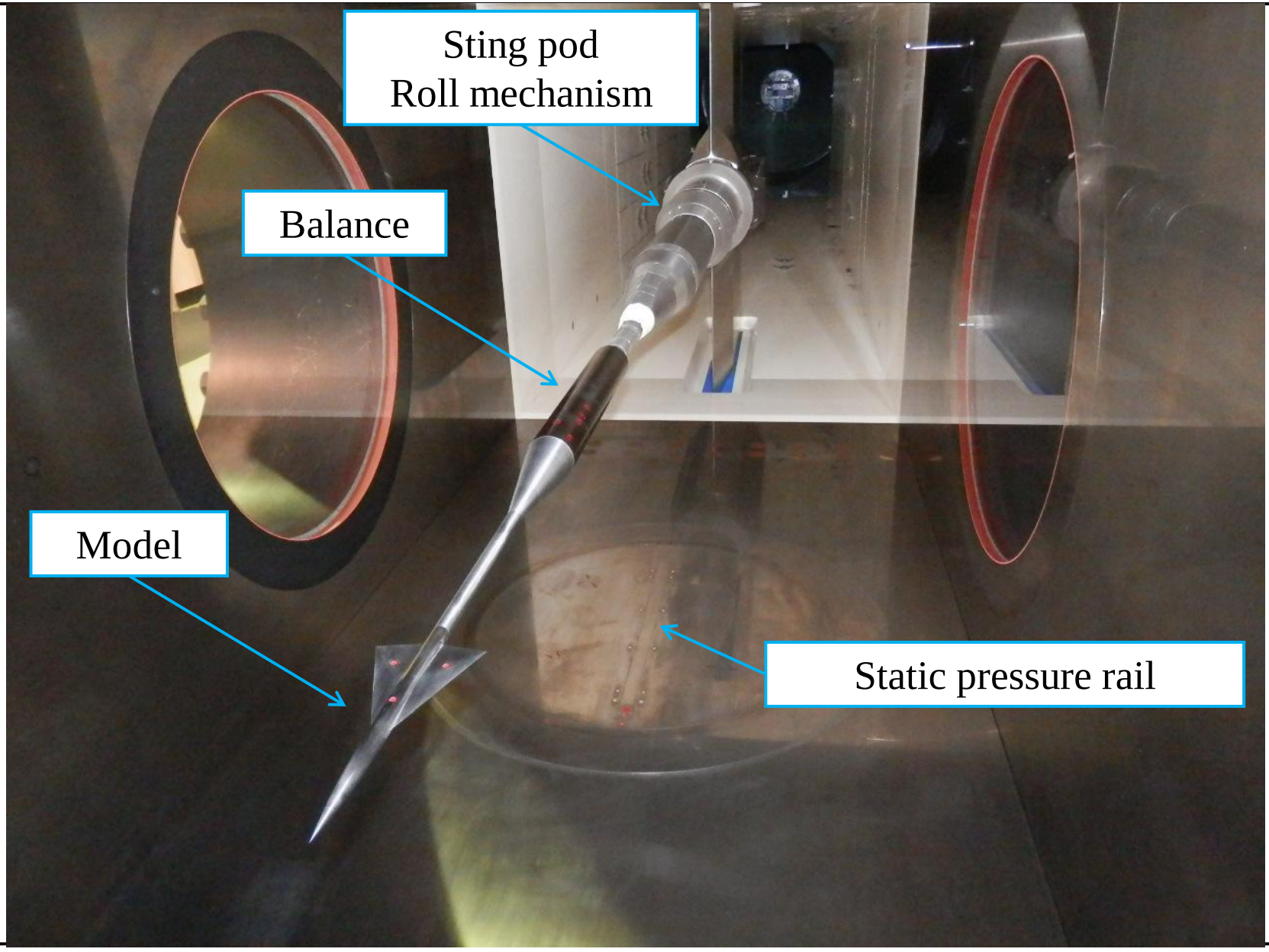
- To obtain validation data for sonic-boom prediction tools.
  - Simple body of revolution models.
  - 69-degree swept-back angle delta wing body model. (for 1st Low Boom Workshop)

JAXA Wind Tunnel Technology Center(WINTEC)
1m x 1m Supersonic Wind Tunnel (JSWT)



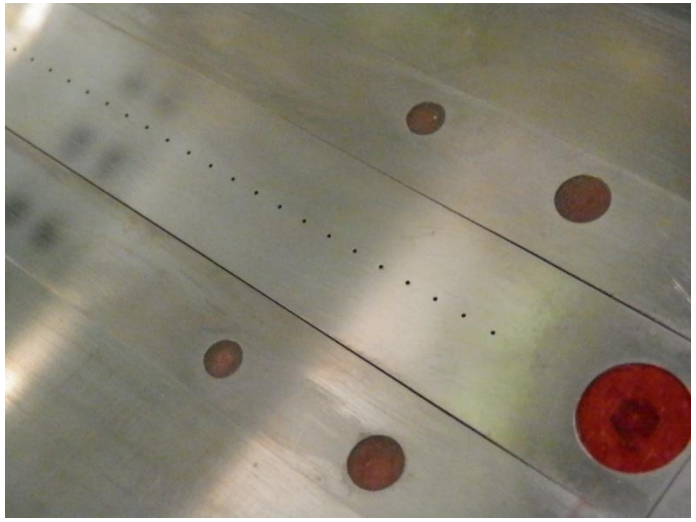
Wind tunnel type	Blow-down type
Test section	1m x 1m
Mach number	1.4 – 4.0
Reynolds number	2 – 6 x 10^7 [1/m]
Mass flow rate	> 280 [kg/s]
Blow duration	< 40 [sec]
Blow interval	about 30 minutes
Built	1961
Refurbished	1999

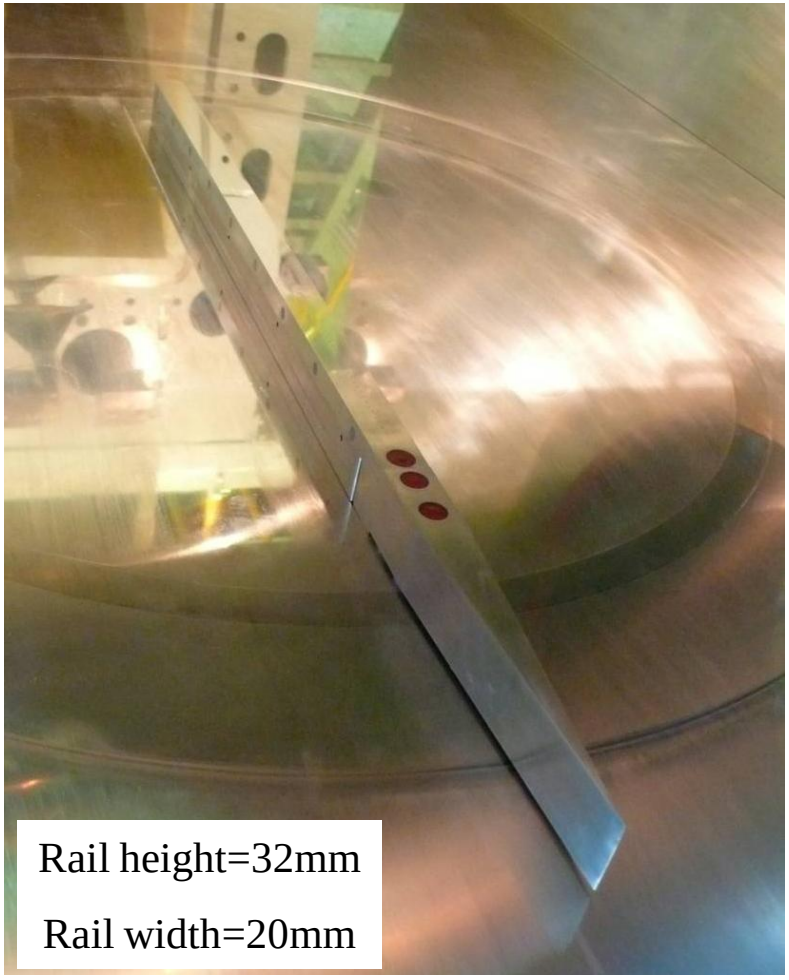
Test Apparatus



Static Pressure Rail

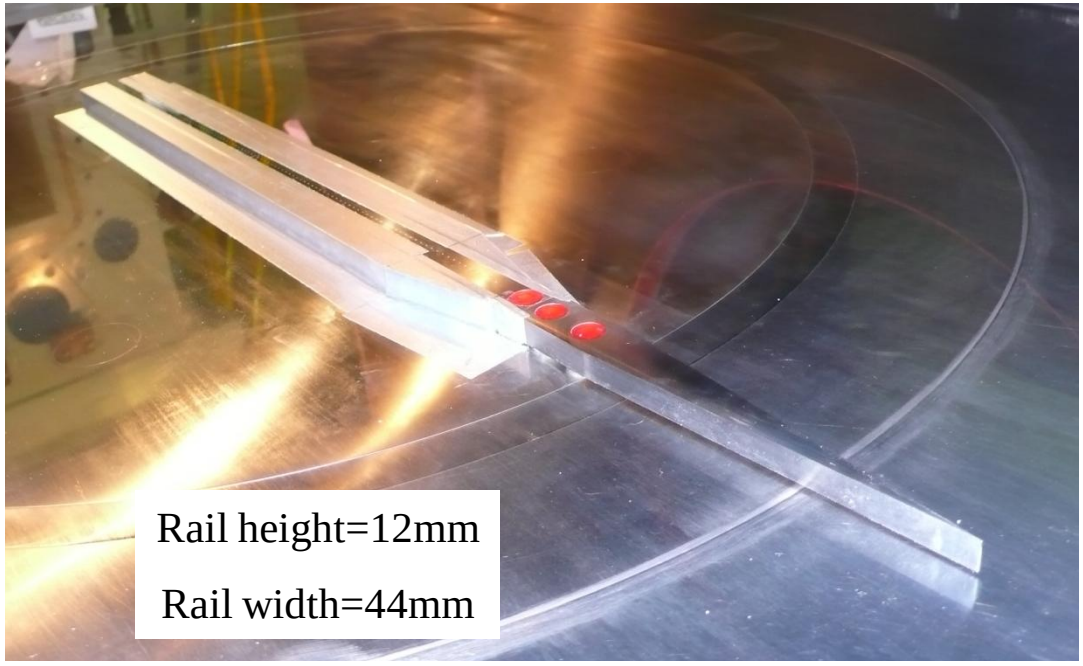
Length=450mm, Width=20, 44mm
Height=0, 12, 32, 52mm
111points pressure taps
(4mm interval, 440mm measurement region)





Rail height=32mm

Rail width=20mm



Rail height=12mm

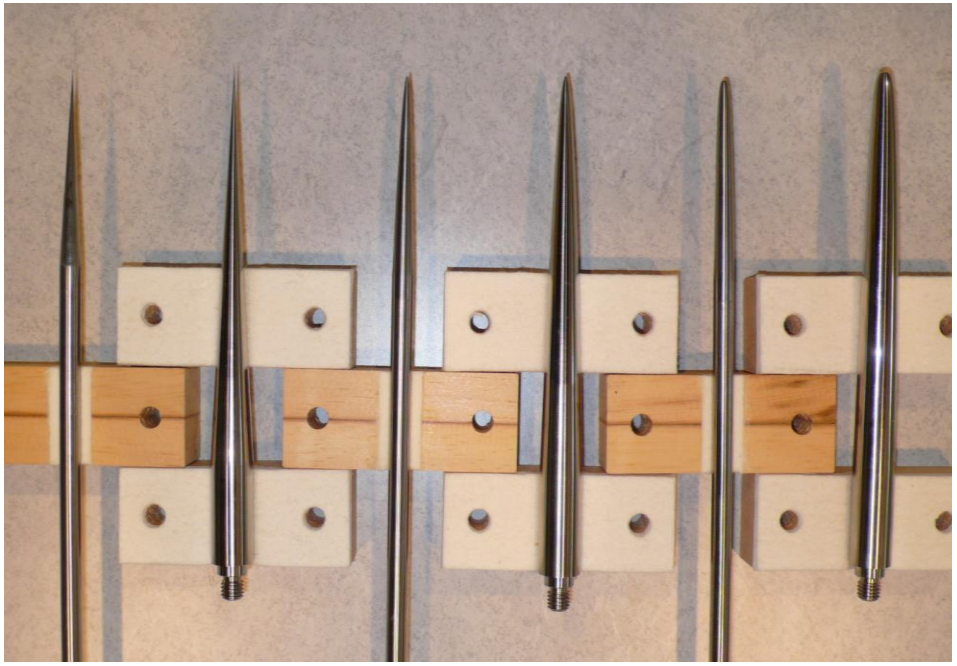
Rail width=44mm

Test Models

Axisymmetrical models

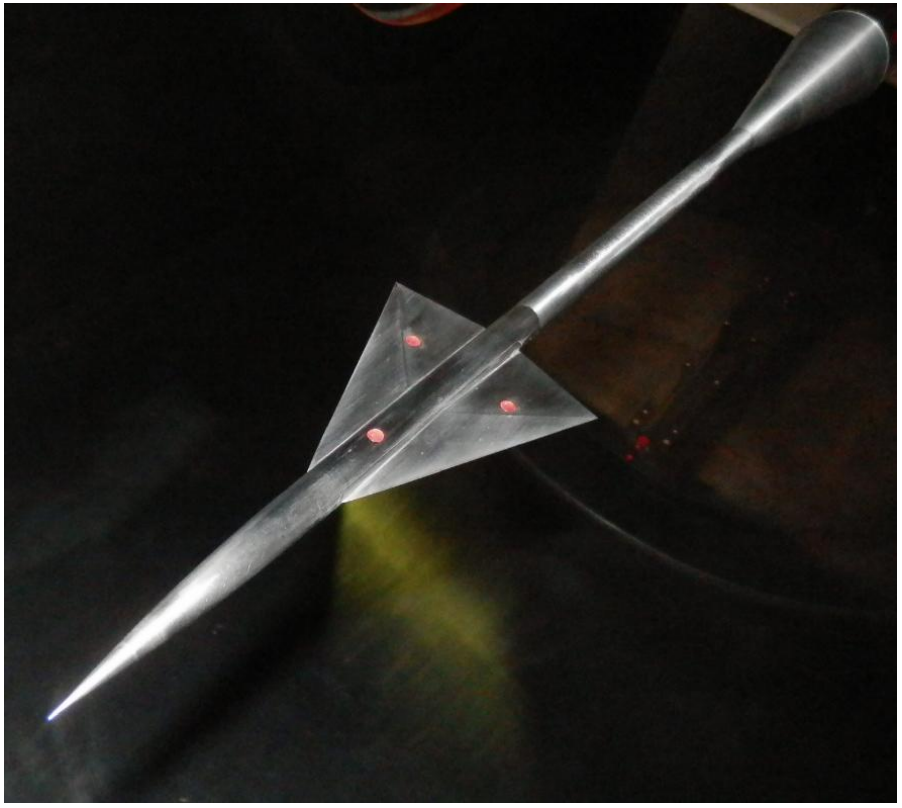
Cone | Parabolic | Quartic

80mm 160mm 80mm 160mm 80mm 160mm



cf. NASA TN D-3160

69-degree swept-back delta wing body model



cf. NASA TN D-7160

Supersonic Wind Tunnel Test in 2011

2011/5/23~6/3

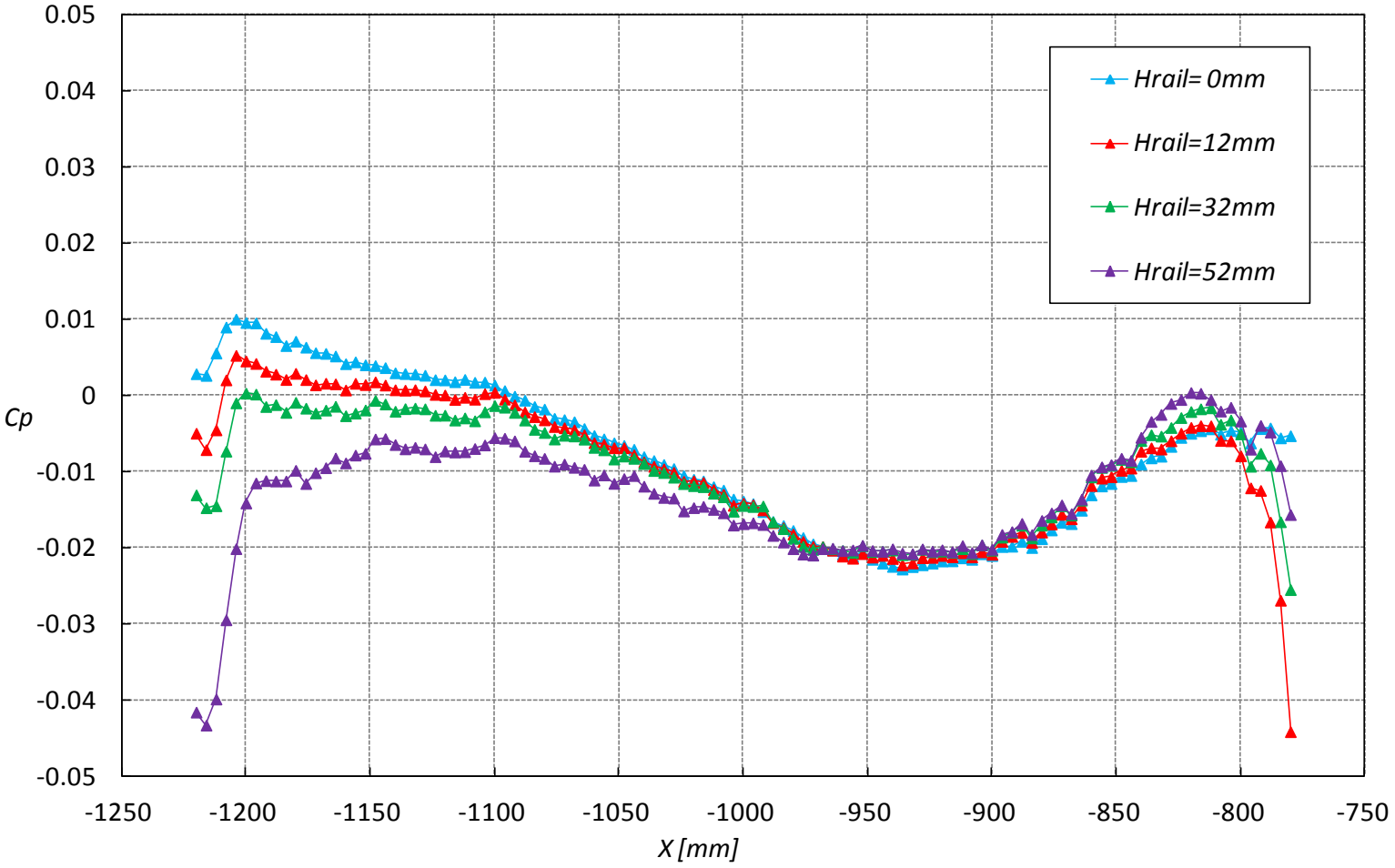
< Axisymmetrical models>

- *To investigate rail height effects*

< 69-degree delta wing body model >

- *To obtain rail data at AoA=0deg (w/o balance)*

Pressure Distribution on the Pressure Rail



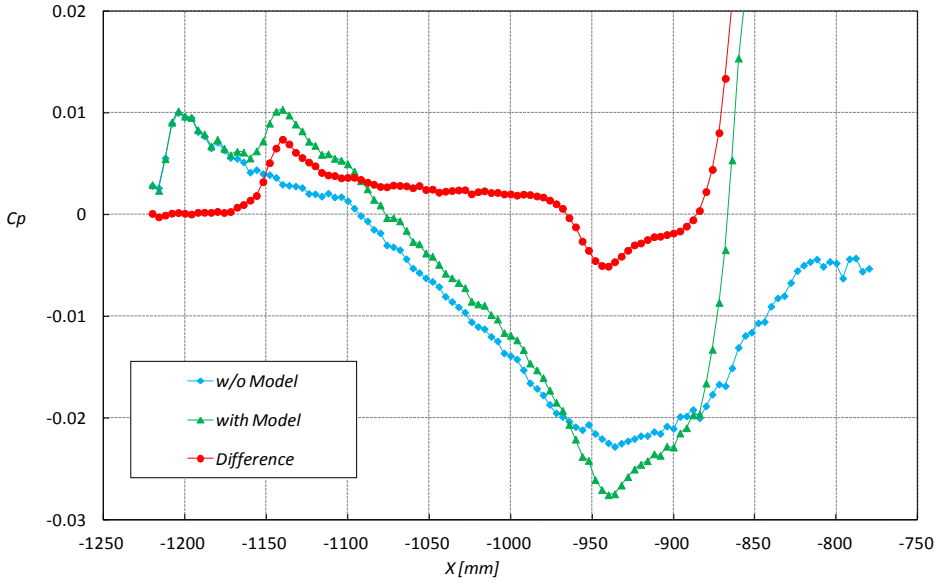
Mach1.4
Q=65kPa
P0=150kPa
Ps=48kPa
Reu=23million
Wrail=20mm

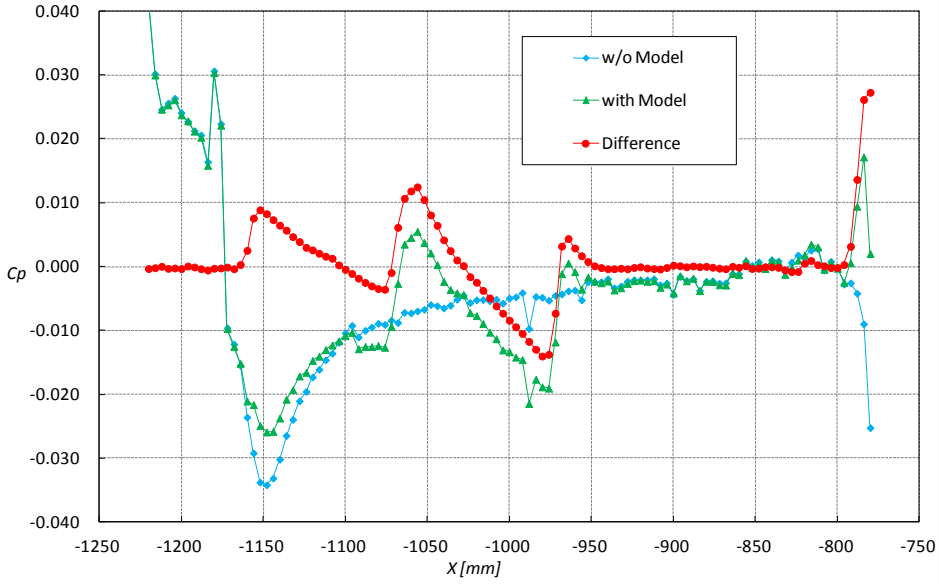
- Some pressure changes are shown on the wind-tunnel wall even without the model.
- The pressure variation gets larger as the rail height increases.

Model On/Off Correction

Parabolic model
M=1.4
α=0deg
H_{rail}=0mm, W_{rail}=20mm

69-degree delta wing body model
M=1.68
α=0deg
H_{rail}=12mm, W_{rail}=44mm

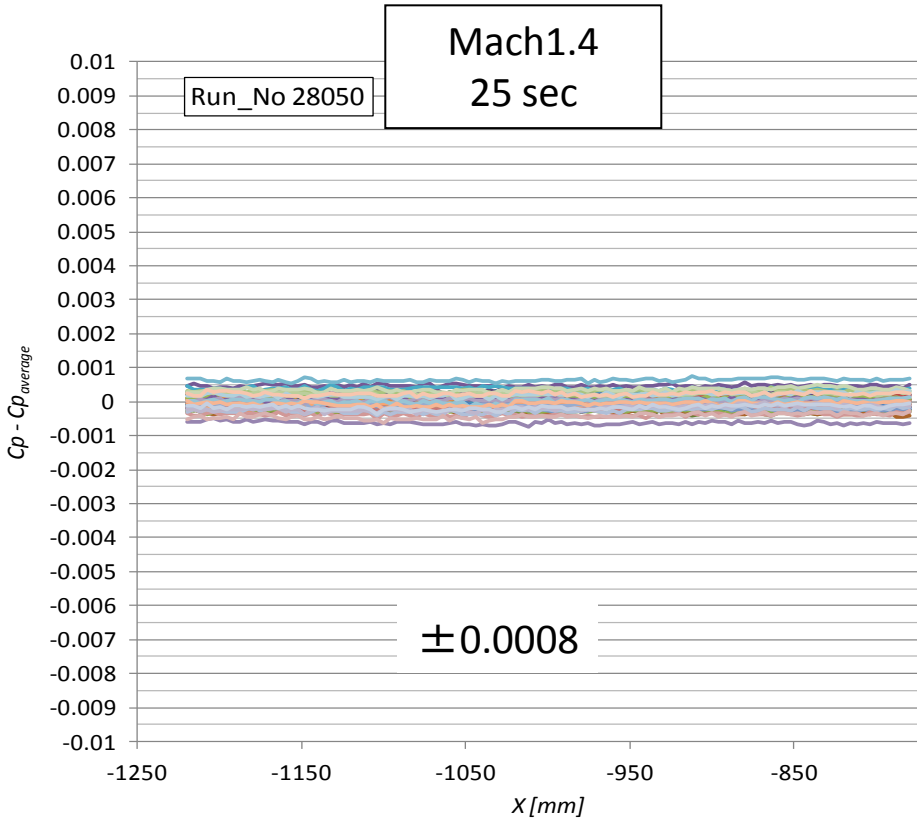


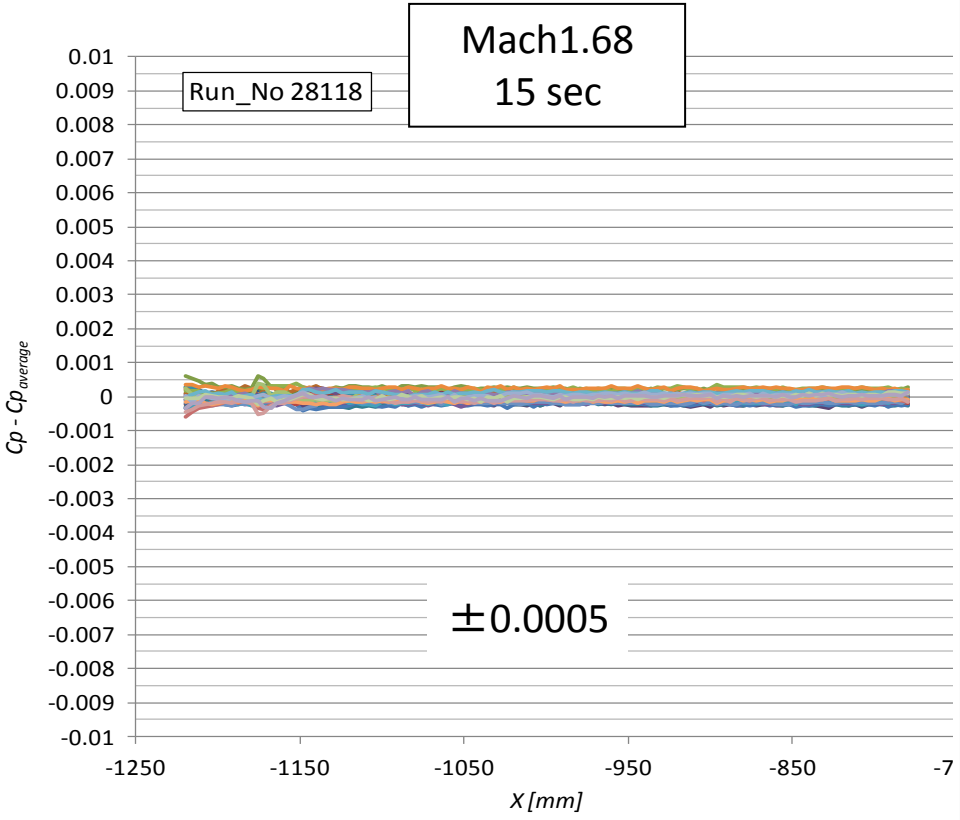


- The result that *Cp* differences at front part of the signatures are almost zero shows the model on/off correction works well in the test.

Pressure Variation(Model Off Case)

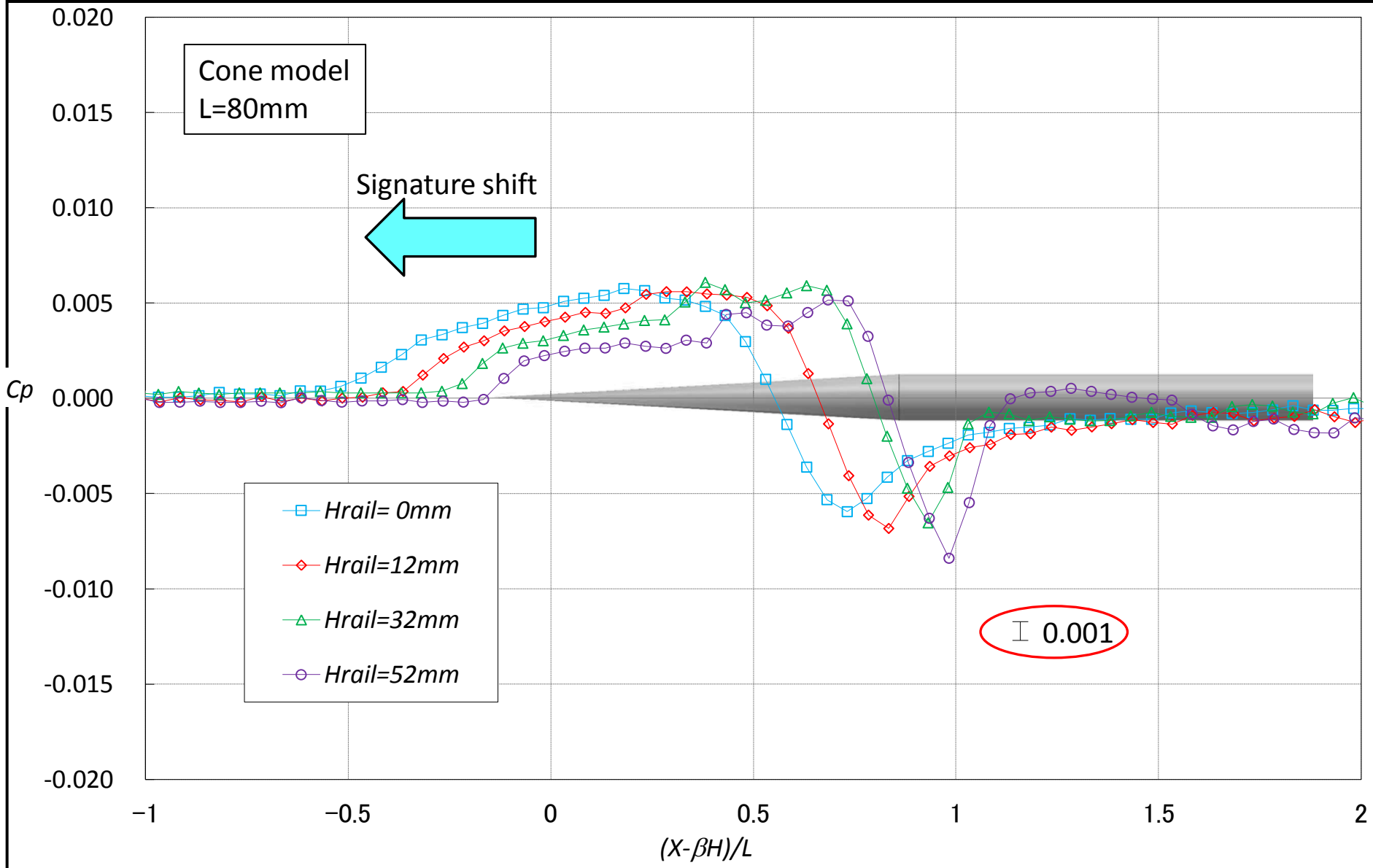
Deviation from the average at each point



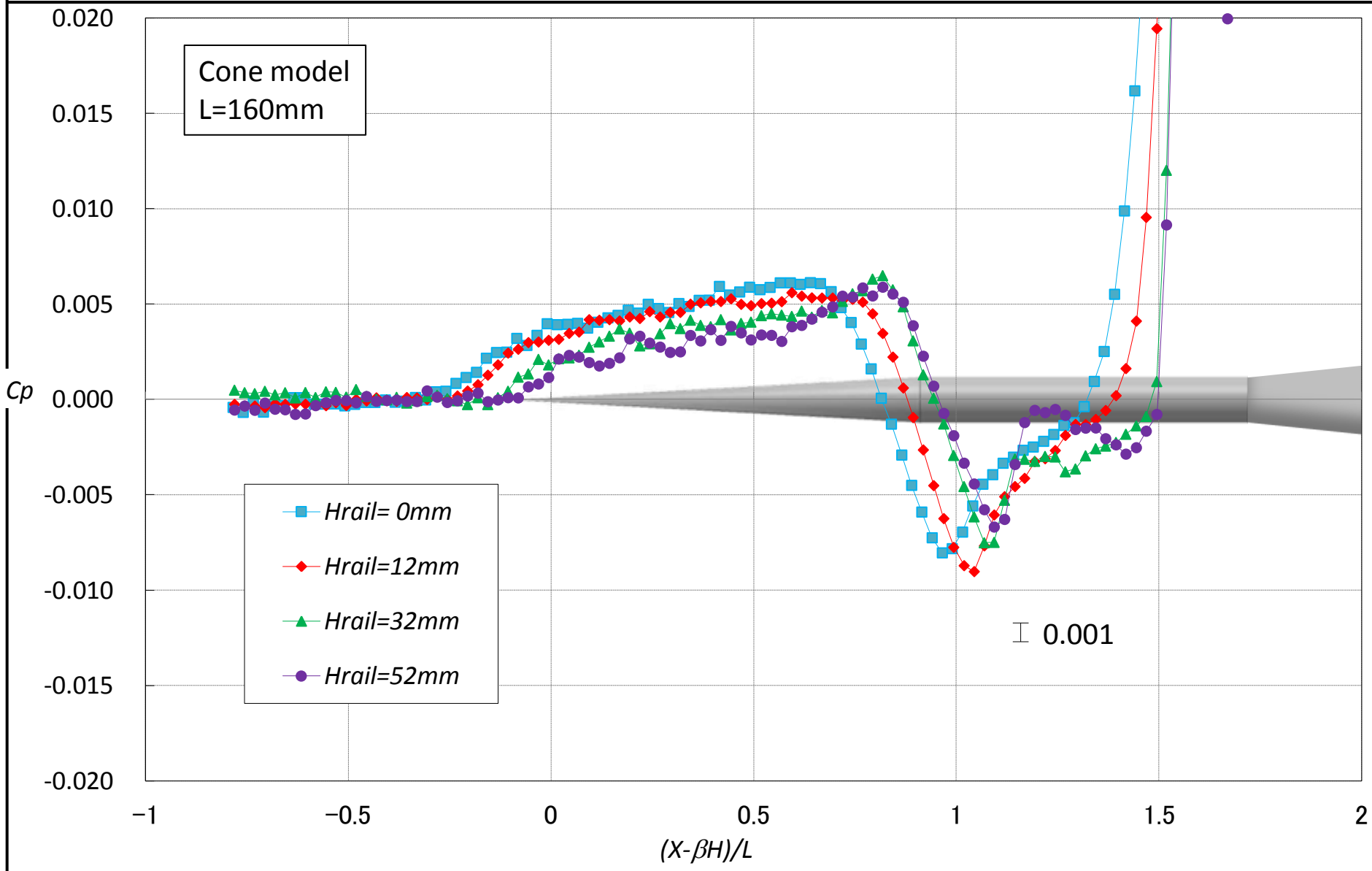


- The *Cp* standard deviations are about 0.00024 for *M*=1.4 and 0.00018 for *M*=1.68 in *Cp*.
- The pressure variation is about 50Pa@M1.4, 35Pa@M1.68 while the accuracy of the used pressure transducer(ZOC 15psi module) is about 160Pa.

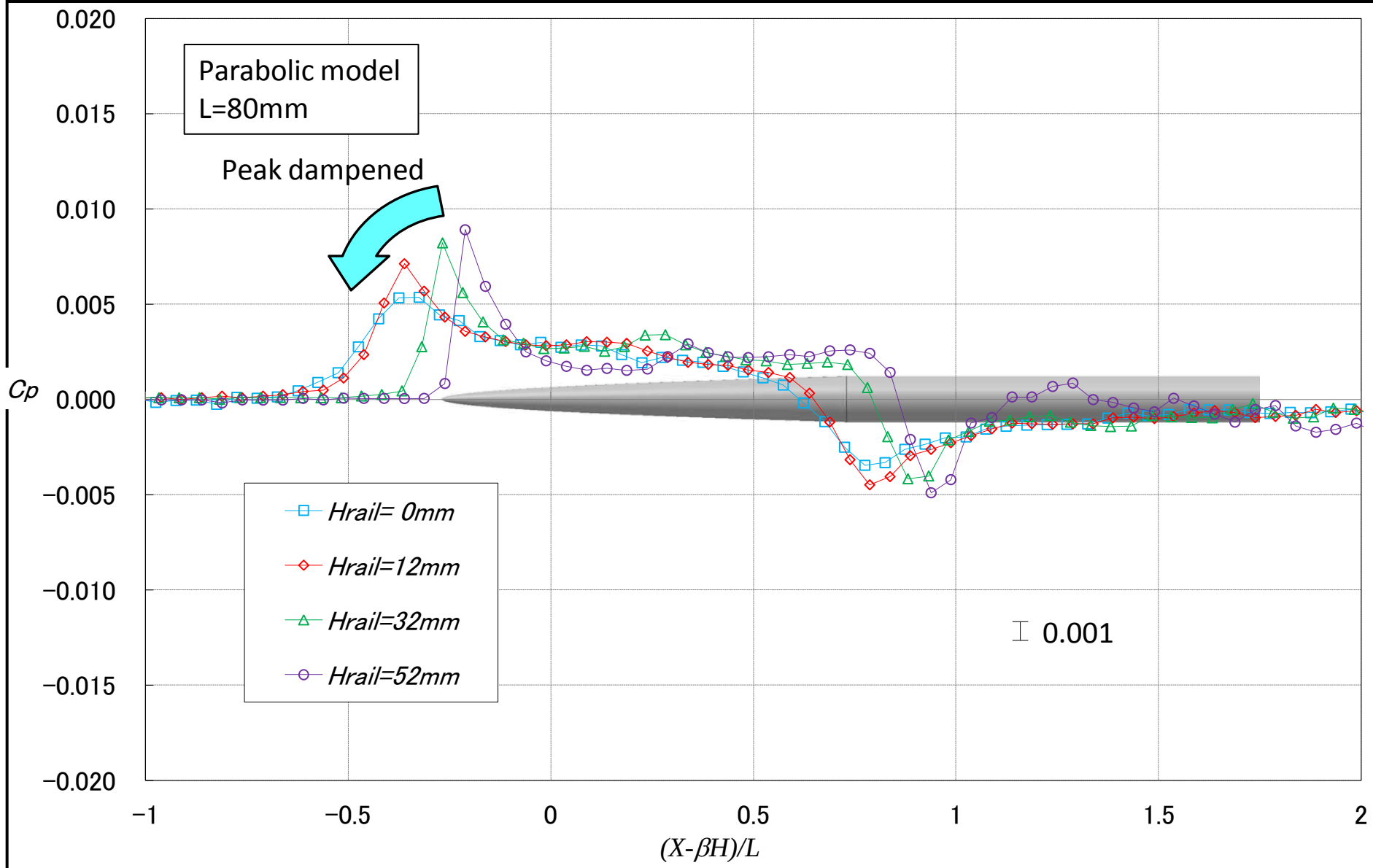
Rail Height Effects



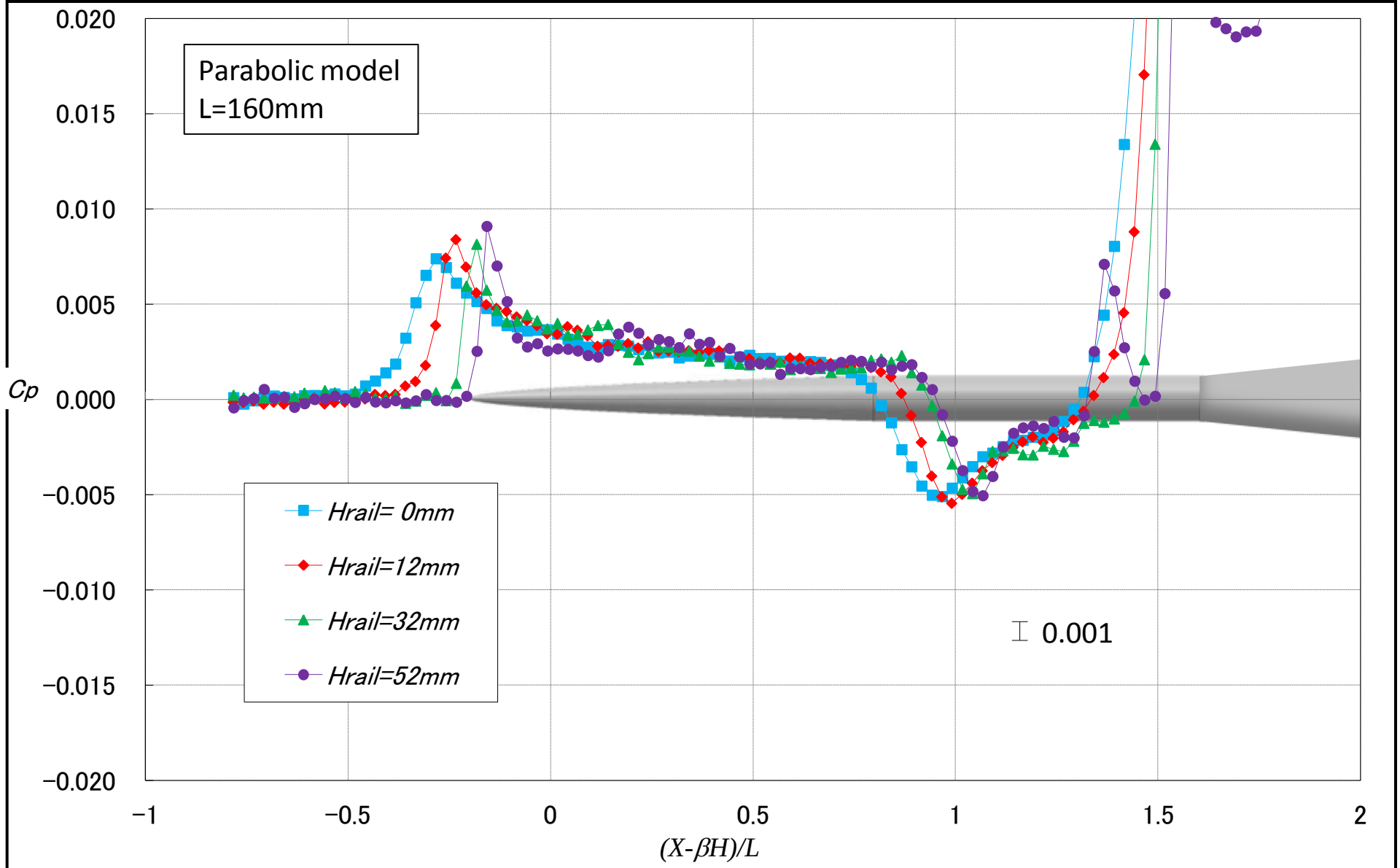
Rail Height Effects



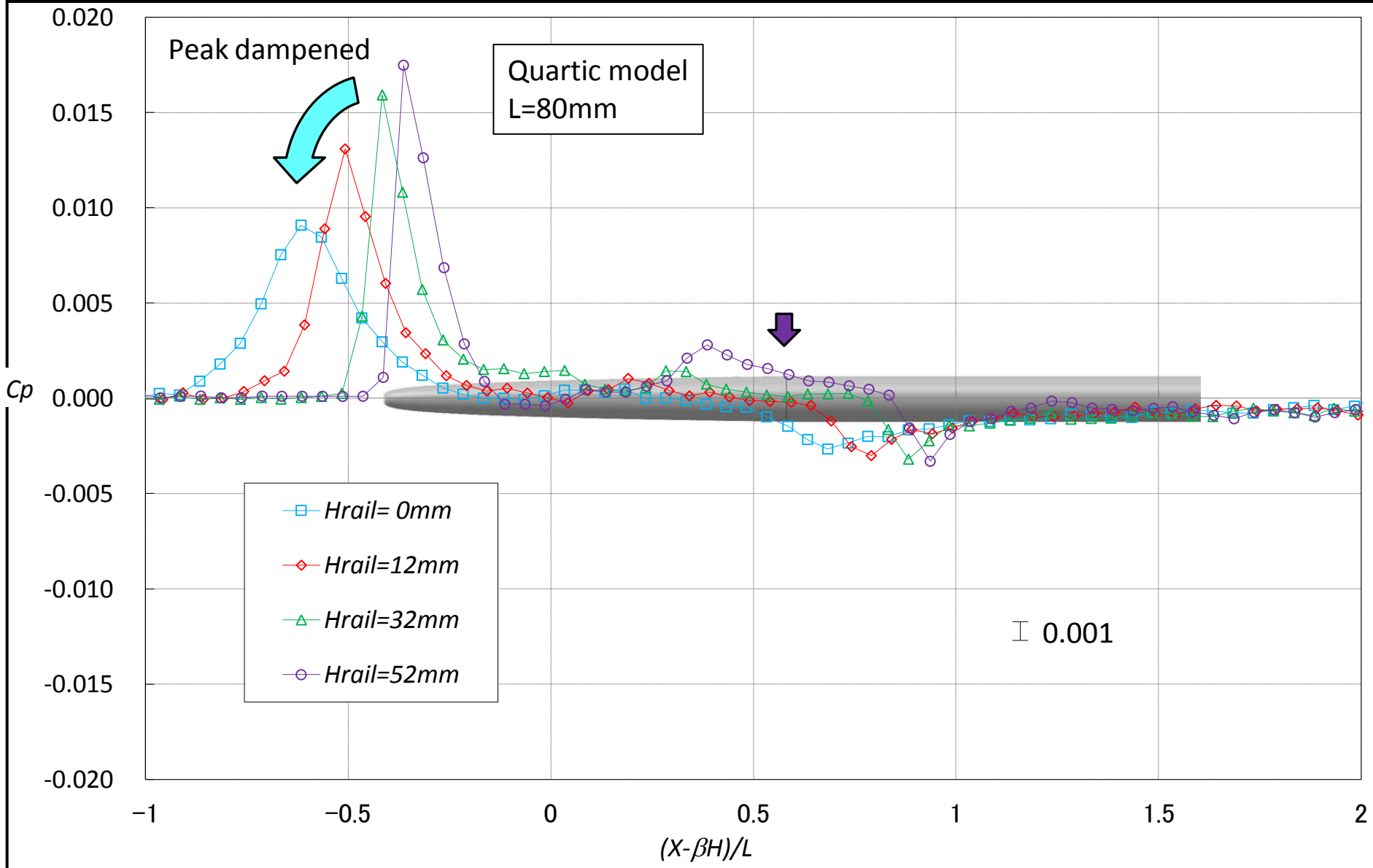
Rail Height Effects



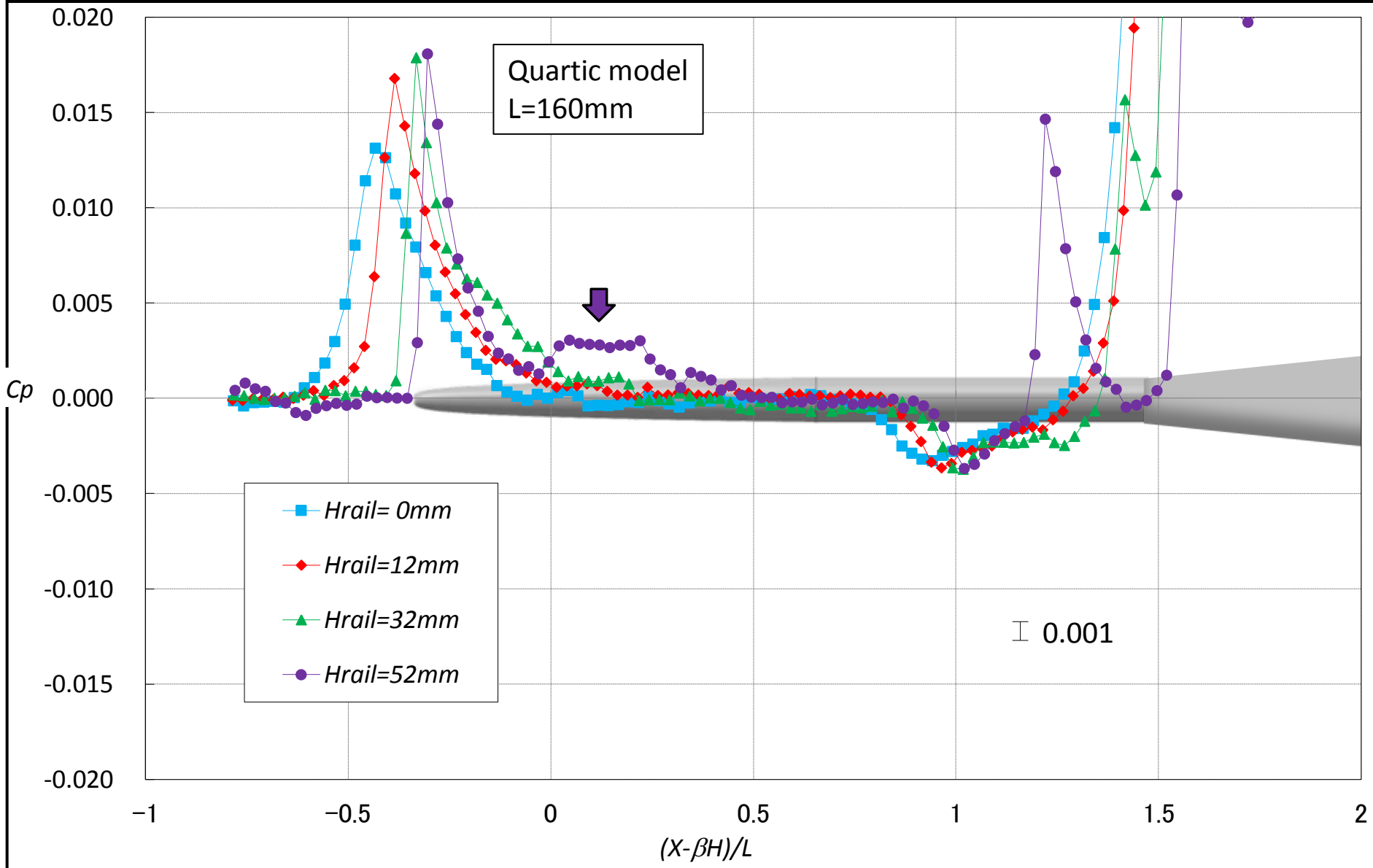
Rail Height Effects



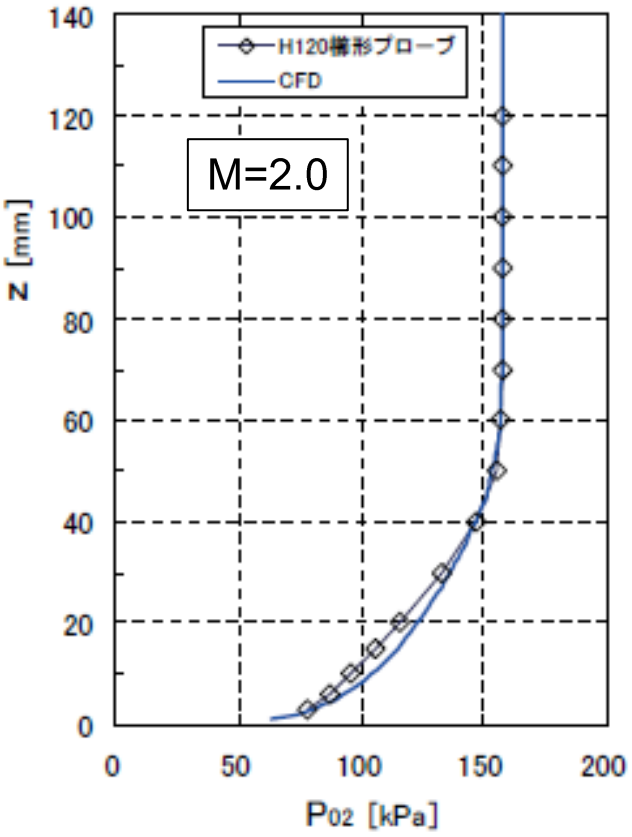
Rail Height Effects

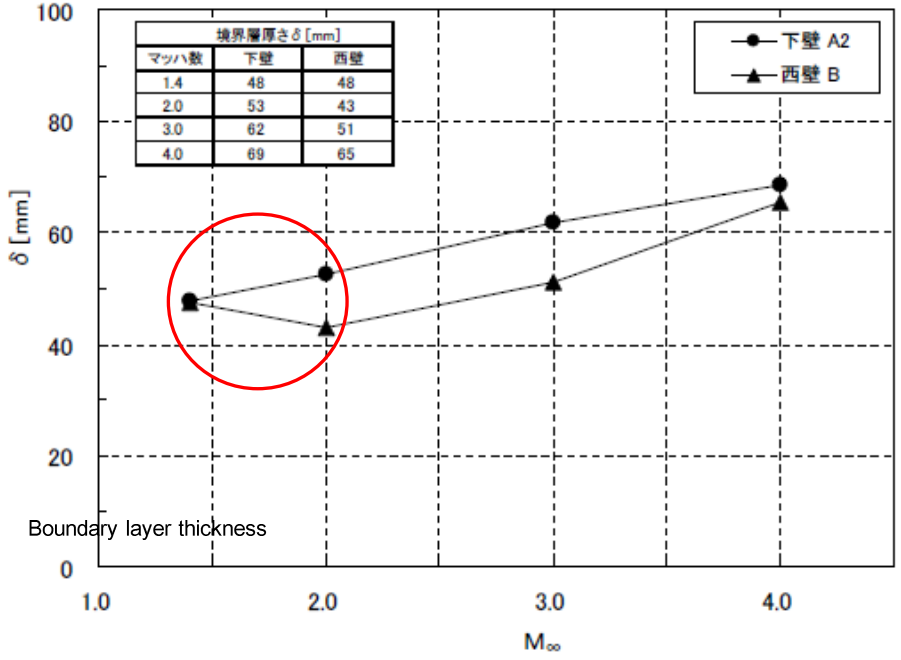


Rail Height Effects

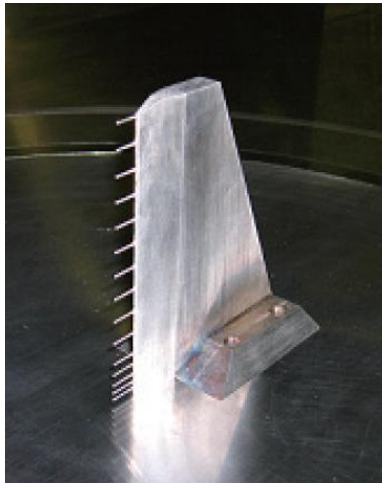


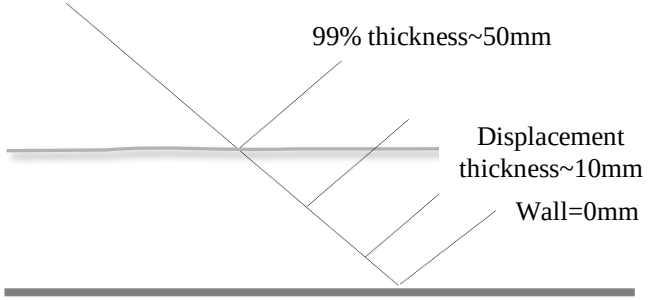
Boundary Layer in JSWT



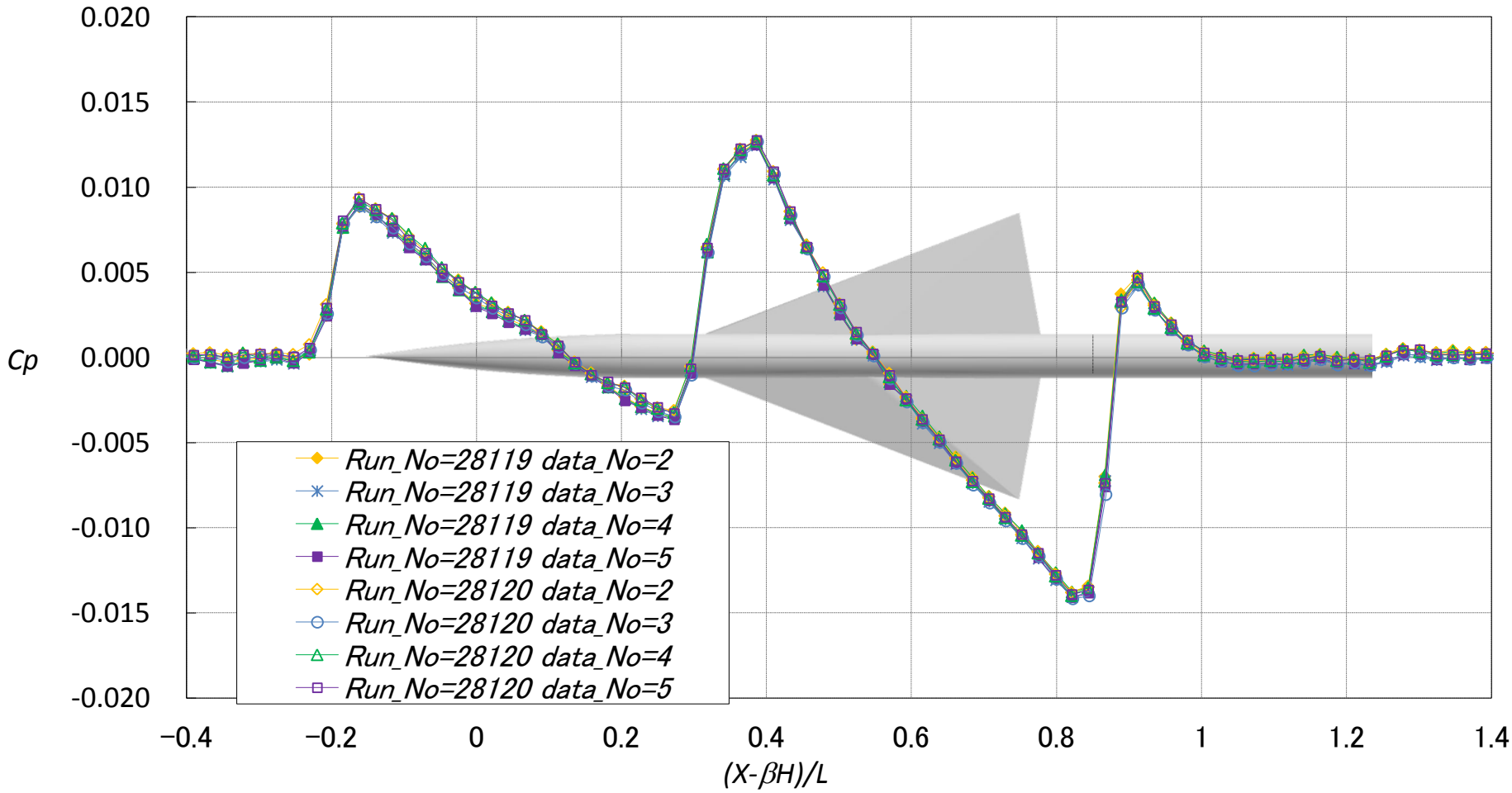


JAXA-RM-08-011



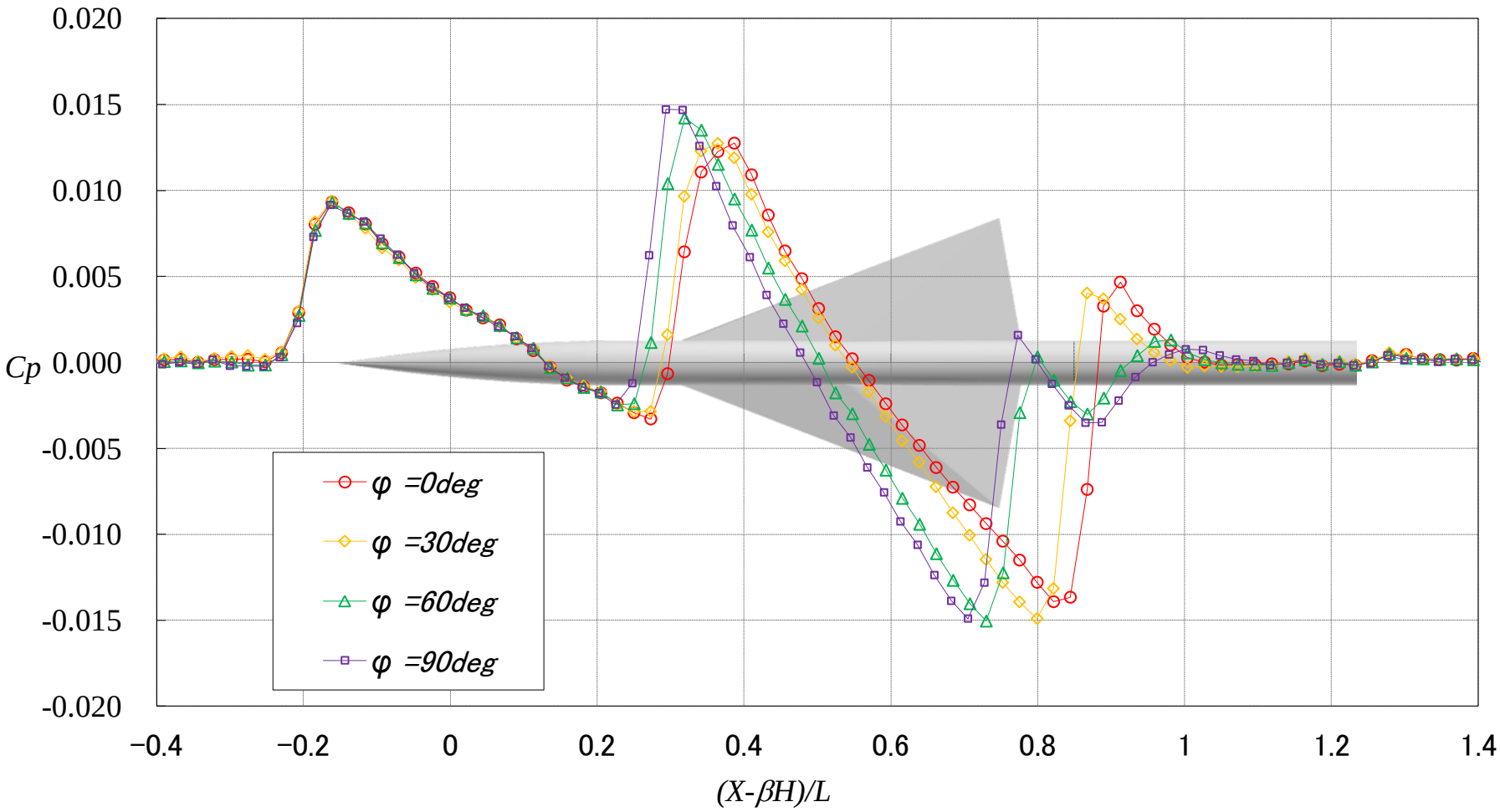


69-degree Delta Wing Body Model



M=1.68, α=0deg, ϕ=0deg, H/L=3.6, H_{rail}=12mm, W_{rail}=44mm

69-degree Delta Wing Body Model



M=1.68, α=0deg, H/L=3.6, H_{rail}=12mm, W_{rail}=44mm

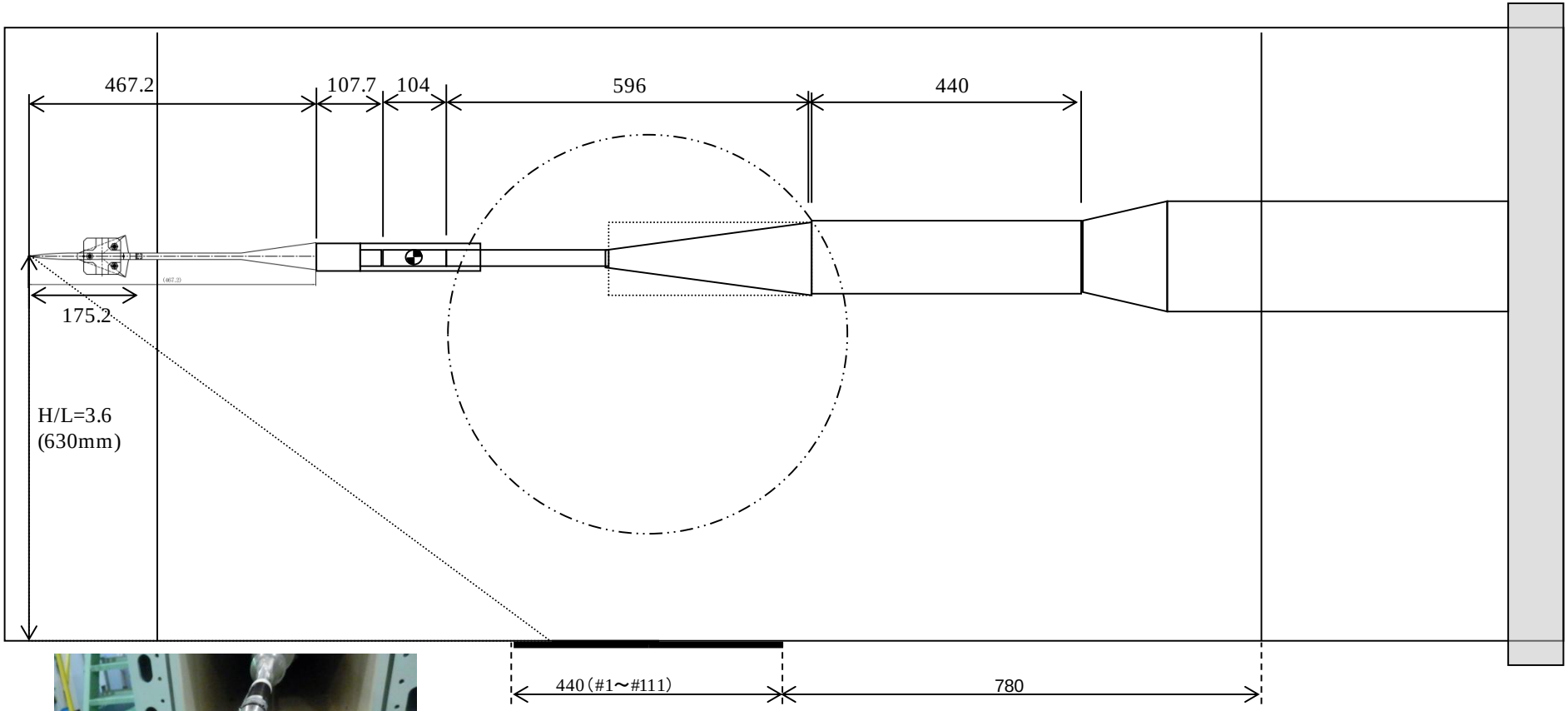
Supersonic Wind Tunnel Test in 2012

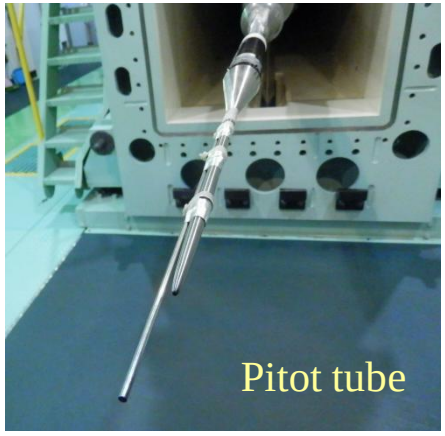
2012/12/26~28

< 69-degree delta wing body model >

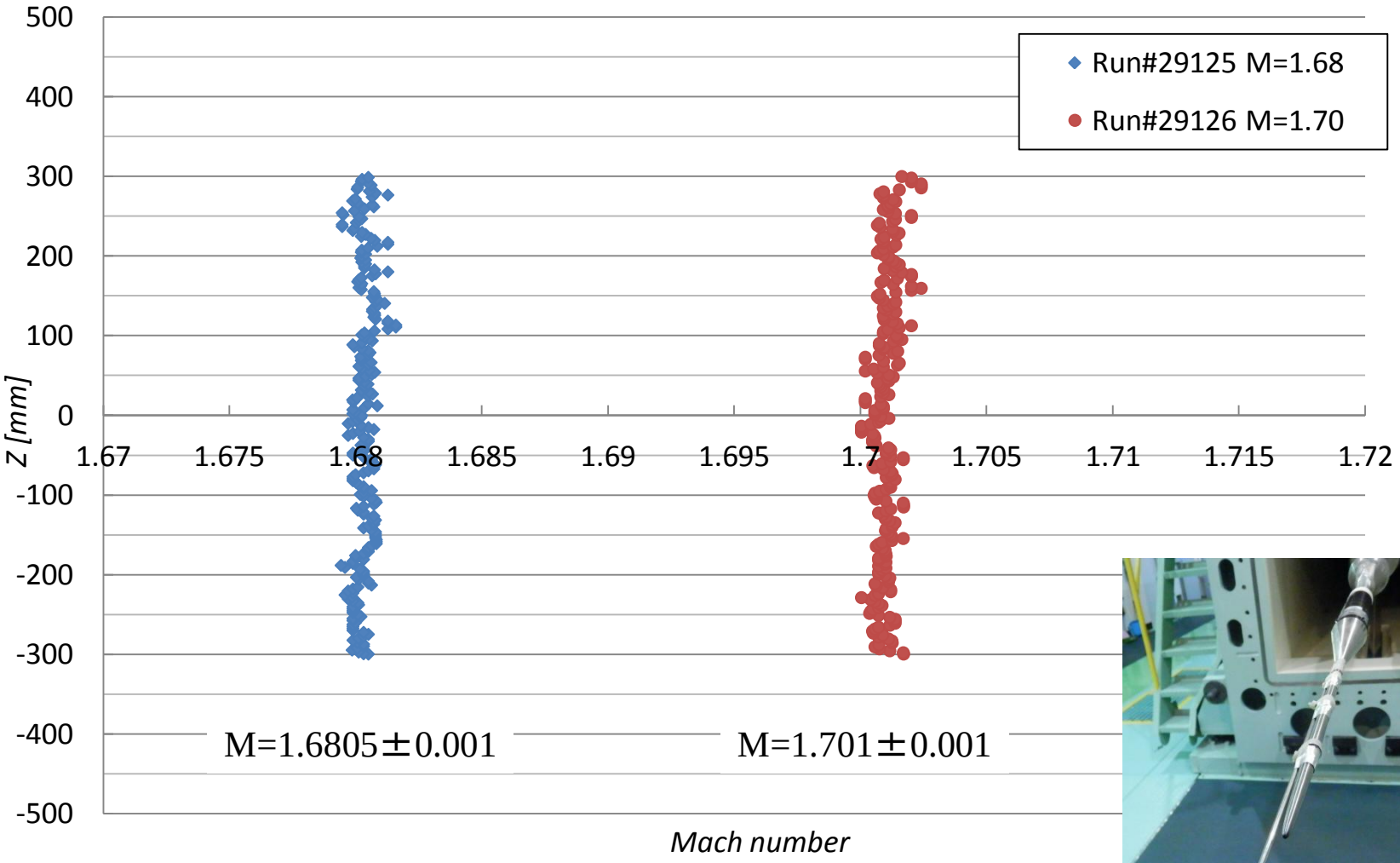
- To obtain rail data at C_L=0.08, 0.15 (with balance)
- To check Mach number distribution
- To check repeatability

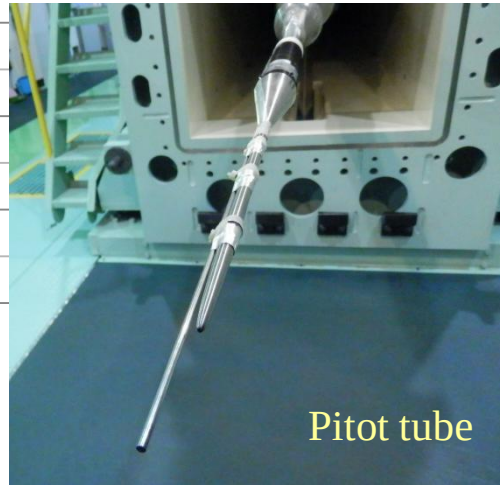
Mach Number Distribution



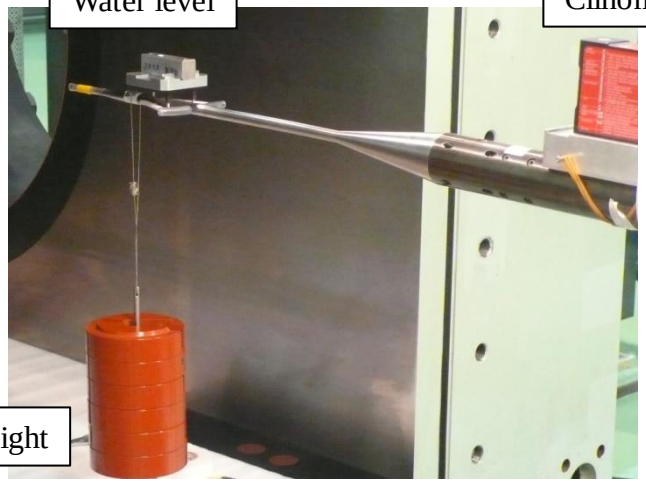


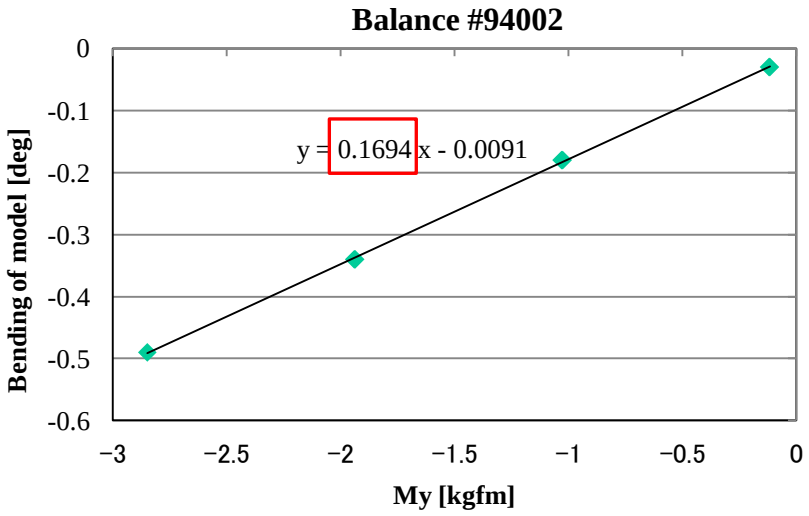
Mach Number Distribution





Model AoA Estimation





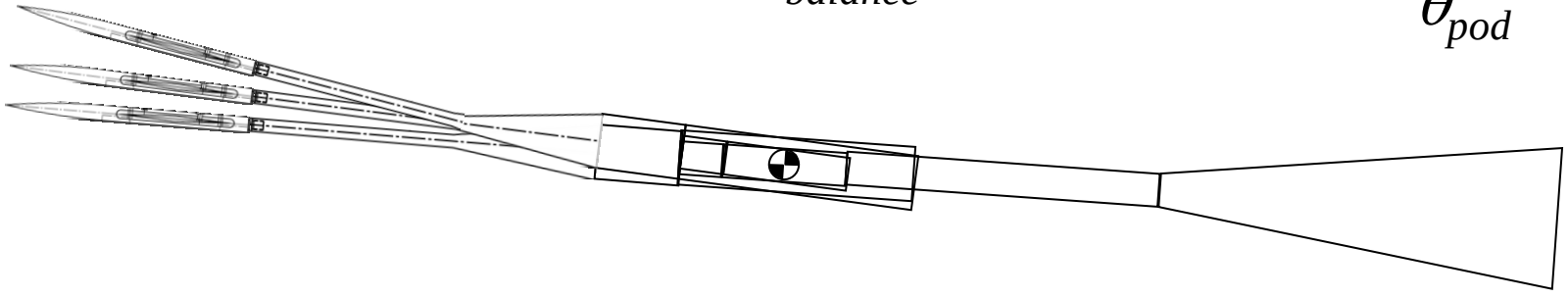
Bending of Model

Bending of Sting&Balance

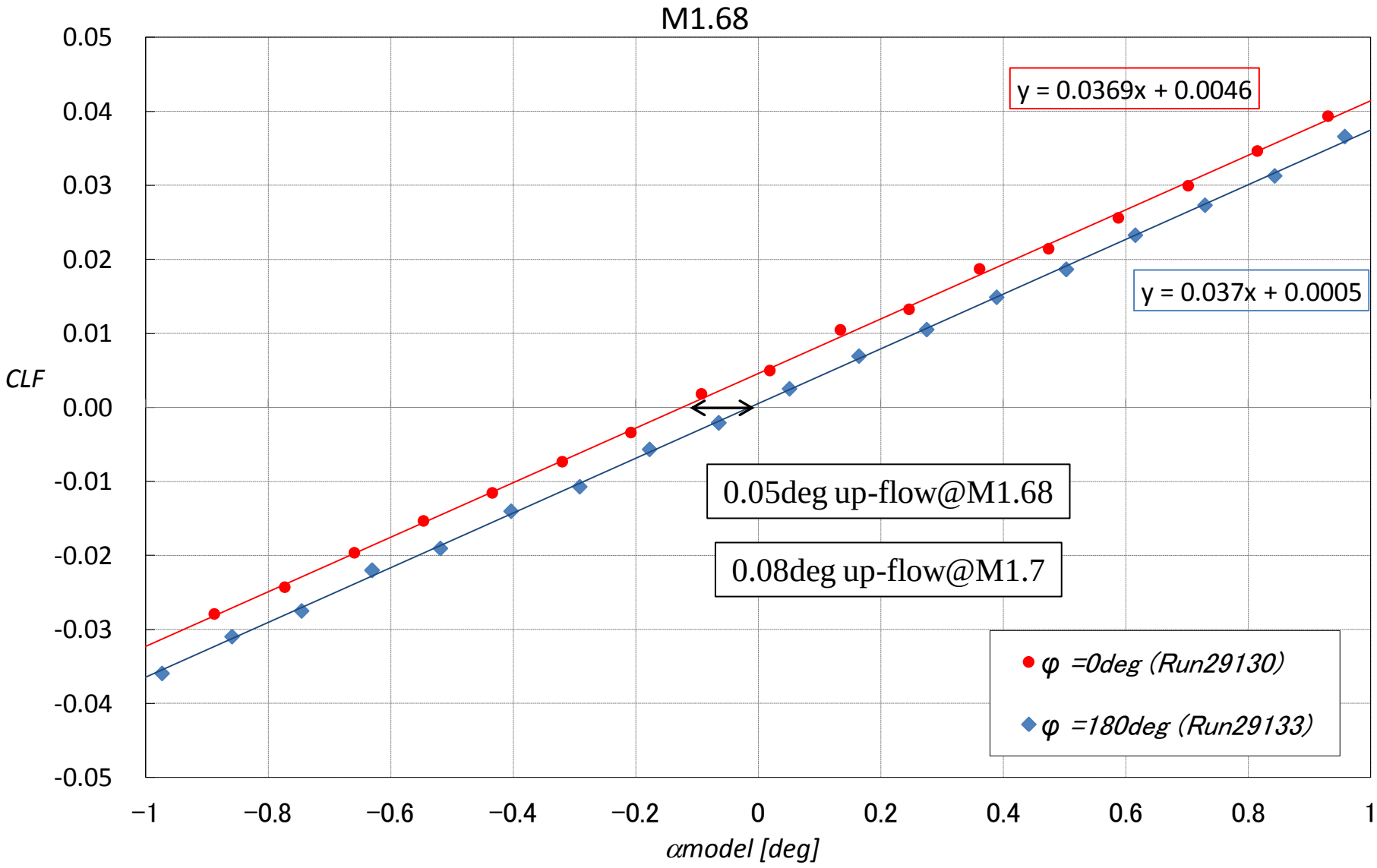
α_{model}

$\alpha_{balance}$

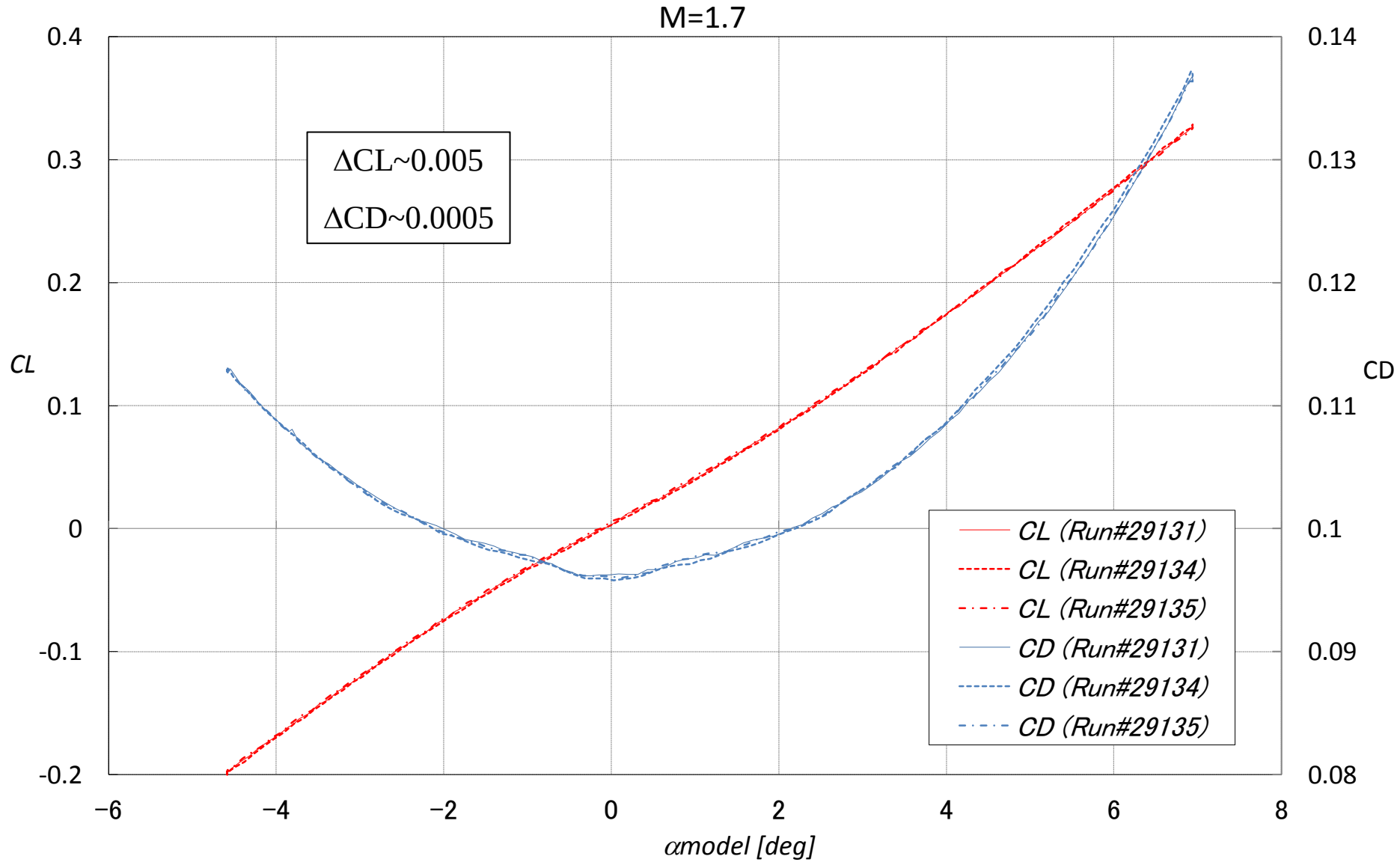
θ_{pod}



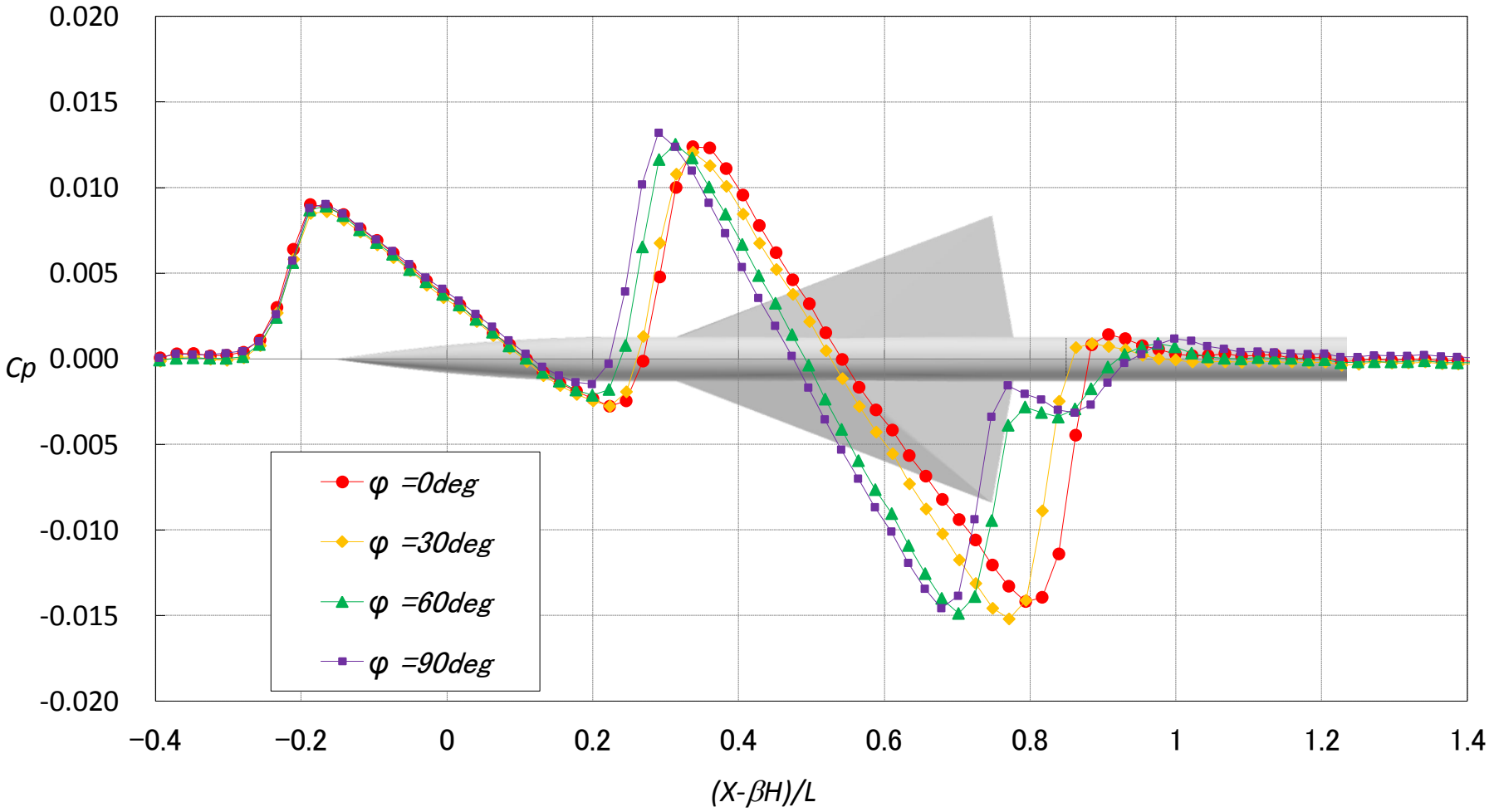
Up-flow/Down-flow



Repeatability

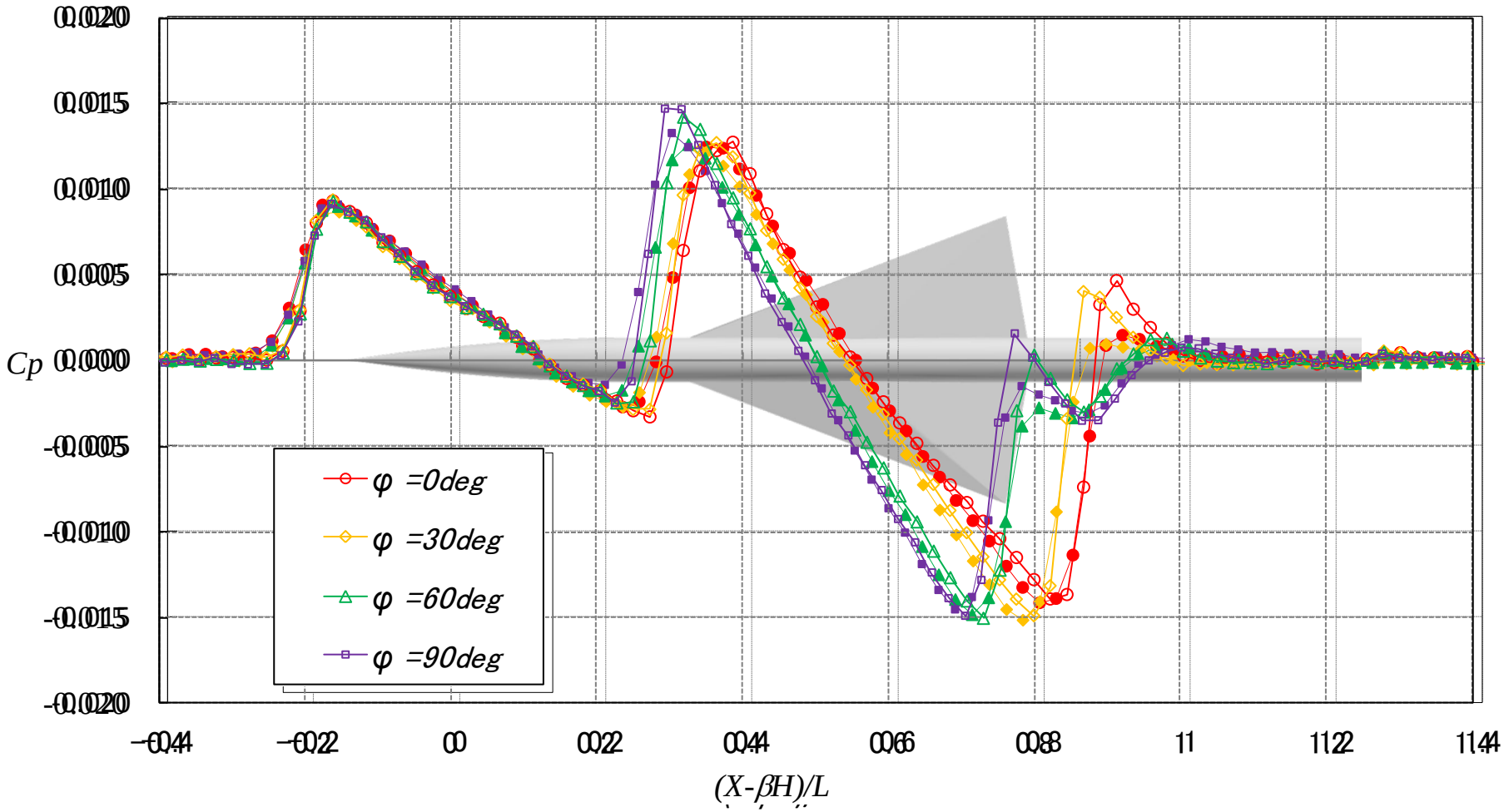


Near-field Pressure Signature for ϕ



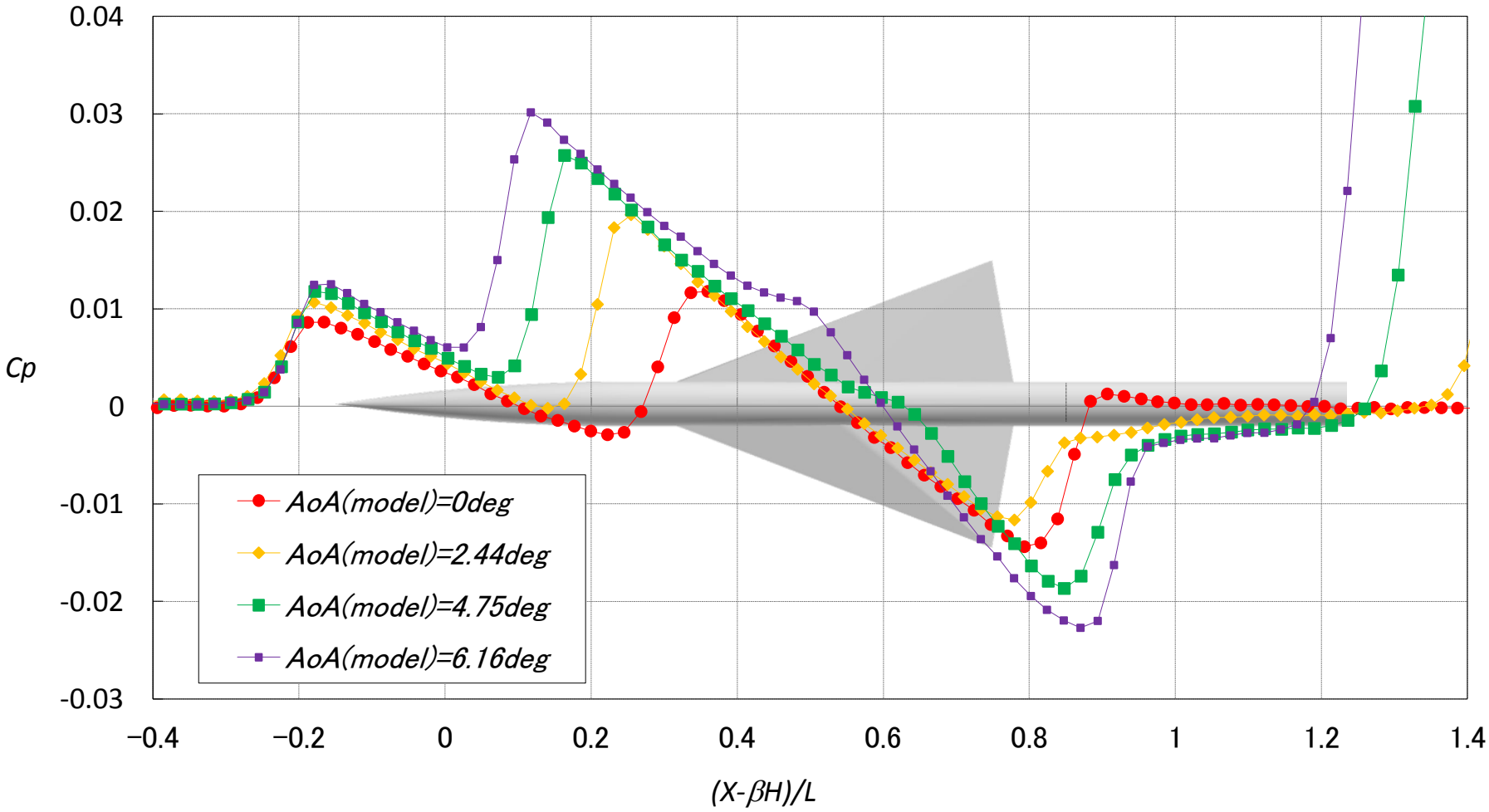
M=1.68, α=0deg, H/L=3.6, H_{rail}=0mm

Comparison between H_{rail}=0mm and 12mm



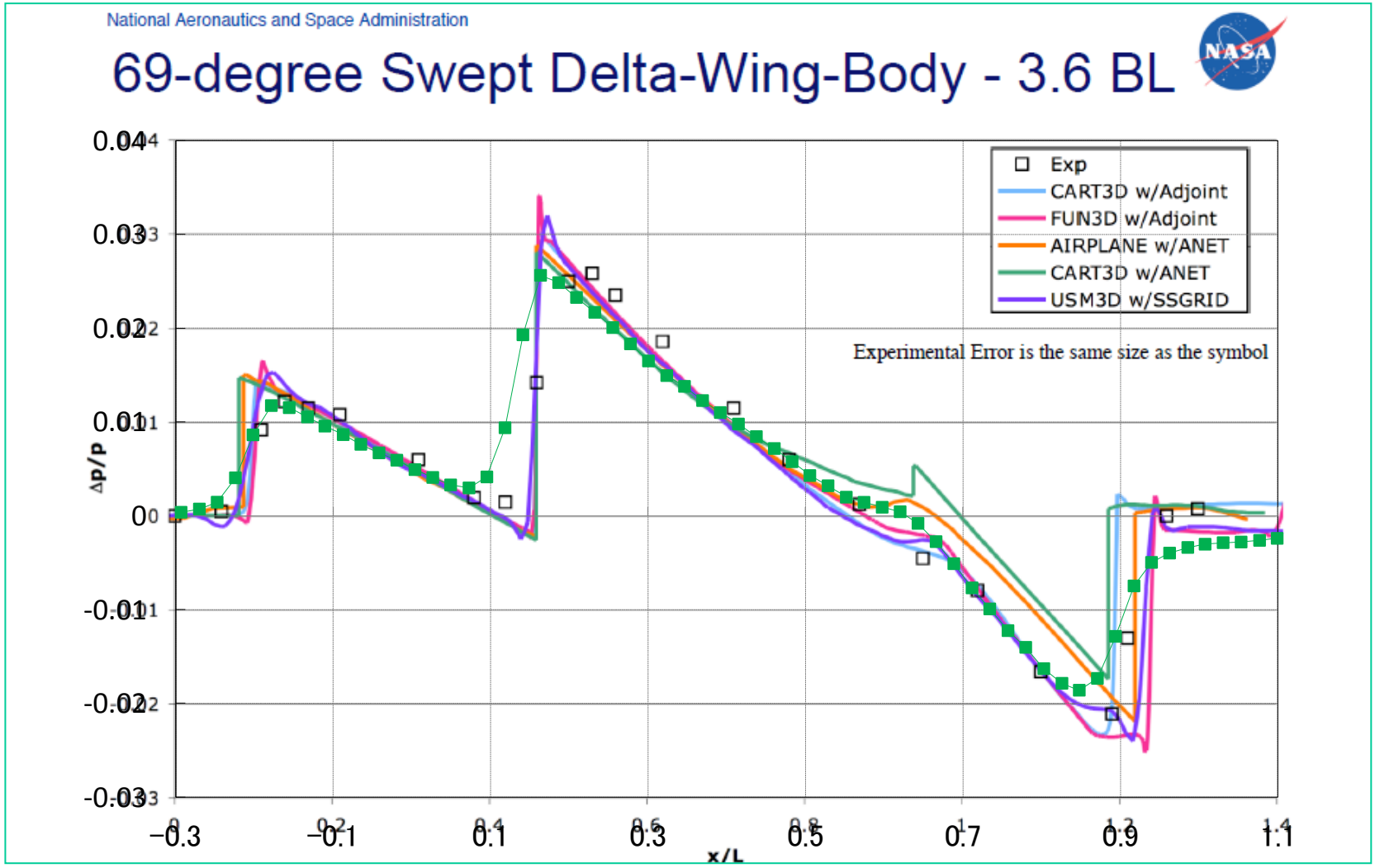
M=1.68, α=0deg, H/L=3.6, H_{rail}=0mm

Near-field Pressure Signature for α



M=1.68, ϕ=0deg, H/L=3.6, H_{rail}=0mm

Comparison with Old NASA Data



Δp/p

(X-βH)/L

Summary

Near-field pressure signatures of several simple models are measured at JAXA's 1m x 1m supersonic wind tunnel(JSWT).

- The pressure measurement accuracy at JSWT using the pressure rail is within 0.0005 in *Cp* at about *M*=1.7.
- The model on/off correction works.
- The shock/boundary layer interaction dampens the measured pressure peaks when the rail height is low.
- The shock/rail interaction distorts the pressure signatures when the rail height is high.
- The near-field pressure signatures of the 69-degree delta wing body model are obtained for qualitative tool validation. (without shock/rail interaction effects)

Thank you !

